The Energy Cost of a Phase Shift

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We prove that a passive causal system without zeros in the upper half-plane cannot shift every frequency of a band $[a, b]$ by the same phase $\theta > 0$ without loss: its mean attenuation over the band is at least $\theta\,(K(\kappa) - E(\kappa))/\kappa$ nepers, $\kappa = (b-a)/(b+a)$, and an explicit system attains this. Delays are free, lossless resonances raise the bound, and zeros can lower it.

1. INTRODUCTION

Does it cost energy to shift the phase of a wave? At a single frequency the answer is no. A delay line of length τ multiplies a wave of angular frequency ω by $e^{i\omega\tau}$, and by choosing τ one obtains any phase at all, without loss.[1] The question becomes interesting only when the same phase shift is asked of every frequency in a band. A delay cannot do that, because its phase grows linearly with frequency. A system that does it moves the oscillations of a pulse under a fixed envelope, which is what one means by a phase shift of a signal rather than of a single sinusoid (Figure 2).

This paper answers the question for that case. It was raised in an earlier paper by one of us with S. N. Thakur [1, 2], where a 1° phase shift of a 5 MHz wave was assigned a delay of 555 ps and an "infinitesimal energy loss" $\Delta E = hf\Delta t$. The delay is right; the energy formula is not (Section 2). It has the units of an action, its numerical value in [1] is off by nine orders of magnitude, and a fixed phase shift changes neither the frequency nor the energy of a wave. What survives is the intuition behind it, that a phase shift has an energy price, and that intuition turns out to be correct in a precise and sharp form once the shift is required on a band.

1.1. The result

We work with causal, time-invariant, passive systems, described by a transfer function H that is analytic and bounded by 1 in the upper half-plane $\mathbb{C}_+$, with real impulse response. With the convention that a pure delay is $e^{i\omega\tau}$, the phase $\varphi(\omega)$ of H is a lag when it is positive, and $\alpha(\omega) = -\log|H(\omega)| \geq 0$ is the attenuation in nepers.[2] We call H *zero-free* when it has no zeros in $\mathbb{C}_+$; this class contains the minimum-phase systems, every system e^{ih} with h a passive response function, and the lossless resonances of Foster [11].[3] Our main theorem (Theorem 6.2) reads as follows.

> *Let H be zero-free and passive, continuous and nonzero near the band $[a, b]$, $0 < a < b$, and suppose that its phase on the band is $\varphi(\omega) = \theta + \tau\omega$ with $\theta \geq 0$ and any $\tau \in \mathbb{R}$. Then*
>
> $$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\mathrm{d}\omega \;\geq\; \theta\,\Lambda, \qquad \Lambda = \frac{K(\kappa) - E(\kappa)}{\kappa}, \tag{1.1}$$
>
> *where $\kappa = (b-a)/(b+a)$ and K, E are the complete elliptic integrals. Equality holds if and only if the system has no response outside the band, in a sense made precise by its Herglotz measure, and it is attained by an explicit system $H_\star$.*

The constant is elementary to estimate. Its power series has positive coefficients and

$$\begin{gathered}\frac{\pi}{4}\,\kappa \;<\; \Lambda \;\leq\; \frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2}, \\ \Lambda \sim \tfrac{1}{2}\log\frac{4b}{a} - 1 \quad (a/b \to 0),\end{gathered} \tag{1.2}$$

so for a narrow band of relative width $\Delta f/f$ a phase shift of one degree costs at least $1 - e^{-2\theta\Lambda}$, about $(\pi^2/720)\,\Delta f/f \approx 0.0137\,\Delta f/f$, of the power at the worst frequency,[4] and the same fraction in the geometric mean

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[1] The factor $e^{i\omega\tau}$ is periodic in τ with period $2\pi/\omega$, so a delay of at most one period gives every phase; at 5 MHz one degree takes 555.6 ps (Section 2).

[2] An amplitude attenuation of one neper is $20/\ln 10 \approx 8.686$ dB, and the power transmission is $|H|^2 = e^{-2\alpha}$.

[3] Foster's reactance theorem: the impedance of a lossless one-port, as a function of $z = \omega$, is an odd Herglotz function, so its reactance increases with frequency between consecutive poles. The corresponding transfer factors e^{ih} have $|H| = 1$ on the real axis off the poles of h.

[4] For a narrow band $\kappa \approx \frac{1}{2}\Delta f/f$, so $\Lambda \approx \frac{\pi}{4}\kappa \approx \frac{\pi}{8}\frac{\Delta f}{f}$; one degree is $\pi/180$ radian, and $1 - e^{-2\theta\Lambda} \approx 2\theta\Lambda = 2\cdot\frac{\pi}{180}\cdot\frac{\pi}{8}\frac{\Delta f}{f} = \frac{\pi^2}{720}\frac{\Delta f}{f}$ (Proposition 10.5(iv)).

over the band (Corollary 6.7 and Section 10.3). For an octave the bound is 0.0415 dB per degree, for a decade 0.143 dB per degree[5] (Table II).

Five features of the theorem deserve to be named at once.

Delay is free. The term $\tau\omega$ does not enter the bound.[6] A system may delay the envelope by any amount; what it pays for is the carrier–envelope offset θ (Proposition 3.3).

The bound is sharp. The extremal system $H_\star$ of Theorem 8.2 is given by an abelian integral,

$$H_\star(z) = \exp\Big(i\tau_\star \int_0^z \frac{s^2-m^2}{R(s)}\,\mathrm{d}s\Big),$$
$$R(s)^2 = (s^2-a^2)(s^2-b^2),$$

with $m^2 = b^2E(k)/K(k)$ and $\tau_\star = 2\theta K(k)/(\pi b)$, $k^2 = 1-a^2/b^2$. It is lossless outside the band, its attenuation vanishes at both band edges and peaks at $\omega = m$, and its phase is exactly θ across the band (Figure 8).

Lossless resonances make things worse. A reactive element resonating outside the band, the classical device for shaping phase, contributes a strictly positive term to the loss (Remark 6.4). A lossless system cannot have a constant phase on any band unless it is ±1 (Corollary 4.6).

Phase leads obey the mirror bound. If $\theta < 0$ the same inequality holds for the mean of α weighted by ab/ω^2 (Theorem 6.6), so the worst frequency of a system whose phase is constant modulo 2π loses at least $\mathrm{dist}(\theta, \pi\mathbb{Z})\,\Lambda$ nepers (Corollary 6.7).

Zeros matter. For $0 < \theta < a\pi/(a+b)$ a single zero on the imaginary axis already lowers the attainable mean attenuation strictly below the least value among all zero-free systems with the same argument (Proposition 9.2 and Corollary 9.3). What the optimal loss is for systems with zeros is open (Problem 9.4).

1.2. How the theorem is proved

The proof rests on a single identity. If $H = \pm e^{ih}$ with h a Herglotz function, then for every band

$$\int_a^b \alpha\,\mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\,p(\omega)\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\mu, \qquad (1.3)$$

where $S = [-b,-a]\cup[a,b]$, μ is the Herglotz measure of h, and

$$p(\omega) = \frac{a^2+b^2-2\omega^2}{\sqrt{(\omega^2-a^2)(b^2-\omega^2)}},$$
$$W = \frac{(\rho-1)^2}{\rho}, \qquad \rho(z) = \sqrt{\frac{z^2-a^2}{z^2-b^2}}.$$

The function W is analytic in $\mathbb{C}_+$, its real part equals -2 on the band and is strictly positive on the rest of the real line, and its imaginary part is p on the band (Figure 3). The identity (1.3) then follows from the Kramers–Kronig relations [6–8] in their Herglotz form. Since $\int_a^b \omega p\,\mathrm{d}\omega = 0$, a delay drops out; since $\int_a^b p\,\mathrm{d}\omega > 0$ and the last term is nonnegative, a positive constant phase forces loss. The value of $\int_a^b p\,\mathrm{d}\omega$ is an elliptic integral, which a Landen transformation brings to the form (1.1).

Sum rules of this kind have a long history. Bode's gain–phase relations [10] bound the phase of a minimum-phase system by its gain; Fano [12] bounded the bandwidth of impedance matching; Rozanov [14] bounded the thickness of absorbers; and Gustafsson, Sjöberg, Bernland and Luger [15, 16] turned the Herglotz representation into a general machine for physical bounds. Miller [17] bounded the delay–bandwidth product of slow light. The inequality (1.1) is, to our knowledge, new: we did not find a bound on the loss forced by a frequency-independent phase shift in this literature, nor in the work on broadband phase shifters, where constant phase *differences* between two lossless channels are realised [13] and constant absolute phase is not attempted. The test function W, which is tailored to an affine phase, is what makes the bound sharp.

1.3. What the result is, and what it is not

The theorem is a sharp inequality of classical analysis with a direct physical reading: a carrier–envelope phase shift over a band has a minimal price in absorbed energy, proportional to the shift and, for narrow bands, to the relative bandwidth. It is not a statement about energy conservation: a constant phase factor is energy neutral (Proposition 3.1), and the price is paid because causality and passivity tie the phase of H on the band to its modulus everywhere. Nor does it rest on quantum mechanics; when the wave is quantised, the bound becomes a lower bound on the probability that a quantum in the band is absorbed (Section 10.3). The energy hf of a quantum, which played the main part in [1], enters only through the kinematics of frequency shifts in Section 3; see also [3–5] for other uses of the photon energy by the first author and coworkers.

1.4. Plan

Section 2 checks the equations of [1]. Section 3 separates the three things a "phase shift" can mean: a constant factor, a delay, and a phase that changes in time. Section 4 sets up passive systems and their Herglotz functions, and Section 5 constructs the test function. The sum rule and the main theorem are in Section 6, the evaluation of the constant in Section 7, the extremal systems in Section 8, and the effect of zeros in Section 9. Section 10 applies the results to waves in passive media and returns to the

[5] These are $\Lambda\cdot\frac{\pi}{180}\cdot\frac{20}{\ln 10}$ with $\Lambda = 0.27353$ for $b/a = 2$ and $\Lambda = 0.94478$ for $b/a = 10$.

[6] The reason is the vanishing first moment $\int_a^b \omega\,p(\omega)\,\mathrm{d}\omega = 0$ of the weight p in (1.3), Lemma 5.3(i).

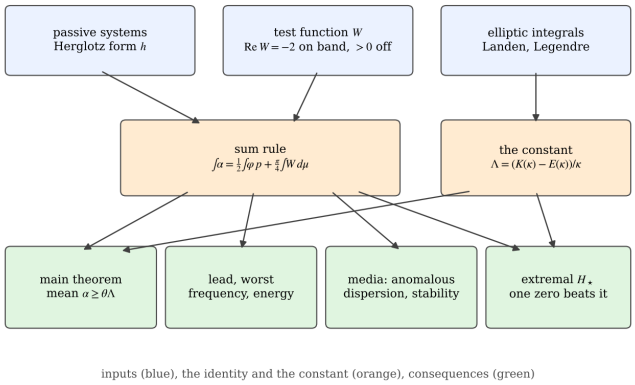


FIG. 1. How the results depend on each other. The sum rule (Theorem 6.1) is built from the test function W (Section 5); the constant Λ comes from the moments of the weight (Section 7); every later statement is a consequence of these two or of the extremal system of Theorem 8.2.

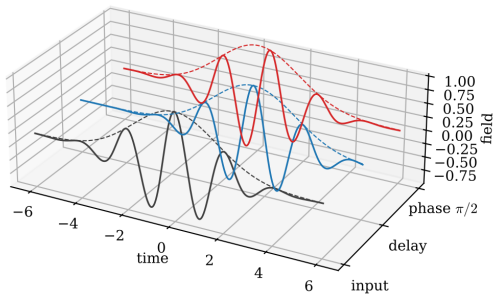


FIG. 2. A pulse (grey), the same pulse after a delay (blue), and after a constant phase shift of $\pi/2$ on its whole band (red). Dashed curves are the envelopes. The delay moves the envelope and the oscillations together; the constant phase shift moves the oscillations under a fixed envelope. Only the second operation costs energy in a passive system.

5 MHz example. Section 11 lists open problems. Figure 1 shows how the results depend on each other. Appendix A derives, from the series alone, every fact on complete elliptic integrals that is used (Legendre's relation, the invariance of the arithmetic–geometric mean and Landen's transformations); Appendix B computes the integrals of the weight; Appendix C derives the formulas for the extremal profile; and Appendix D describes what has been checked by machine. Every statement is proved in the text. The finite and algebraic parts of the proofs (dimensional analysis, the coefficient recurrences, enclosures of Λ and the polynomial identities) are certified in Lean 4, and a script re-evaluates every numerical value.

2. THE EQUATIONS OF THE EARLIER PAPER

We first go through the formulas of [1], since the rest of the paper replaces them. They fall into two groups: kinematic relations between phase, time and frequency, which are correct, and formulas for an energy loss ΔE, which are not.

2.1. Phase, time and frequency

For a sinusoid of frequency f, a delay Δt produces the phase

$$\Phi^\circ = 360^\circ f\,\Delta t, \tag{2.1}$$

and [1] rewrites this as $\Delta t = \Phi^\circ/(360^\circ f)$ and $f = \Phi^\circ/(360^\circ \Delta t)$. These are correct, with the understanding that the last one gives the frequency at which a *given* delay produces a *given* phase; it does not describe a change of frequency. The delay that produces one degree is

$$T(\mathrm{deg}) = \frac{1}{360 f},$$

which at $f = 5\,\mathrm{MHz}$ is 555.6 ps, as stated in [1]. Two of the phrasings there should be read with care. The equation "1° phase shift $= T/360$" equates an angle with a time; what is meant is that a delay of $T/360$ produces a phase of 1°, where $T = 1/f$ is the period. The name "time distortion" for Δt suggests something more than a delay; the quantity in (2.1) is just the delay. Finally, the statement that $T(\mathrm{deg}) \propto 1/f$ is true of the delay needed for one degree, and it is the reason why a delay cannot produce the same phase at two frequencies.

2.2. The energy formulas

The paper [1] proposes three formulas for an "infinitesimal loss of wave energy",

$$\Delta E = h f \Delta t, \qquad \Delta E = \frac{2\pi h f_1}{360}\,T(\mathrm{deg}), \qquad \Delta E = \frac{2\pi h}{360}\,\frac{T(\mathrm{deg})}{\Delta t}, \tag{2.2}$$

with h Planck's constant, and evaluates the first one for $f = 4{,}986{,}111.11\,\mathrm{Hz}$ and $\Delta t = 555\,\mathrm{ps}$. Four things go wrong.

Units. Planck's constant is an action, measured in J s; hf is an energy and $hf\Delta t$ is again an action. The same holds for the other two formulas in (2.2), in which $T(\mathrm{deg})/\Delta t$ is a pure number and $f_1 T(\mathrm{deg})$ is the number $1/360$. None of the three expressions is an energy.[7]

[7] If $T(\mathrm{deg})$ is read as the angle $1^\circ = \pi/180$ rather than as a time, the second formula has units of energy, but it then reads $\Delta E = (\pi/180)\,hf_1\,(2\pi/360)$, a fixed multiple of hf_1 without any connection to the delay.

Arithmetic. The product evaluated in [1] is

$$(6.626\times10^{-34})(4.986\times10^{6})(555\times10^{-12}) = 1.834\times10^{-36},$$

in J s, not 1.848×10^{-27} J. The stated value is larger by a factor of about 10^9, and it would amount to 56% of the energy $E_1 = hf_1 = 3.313\times10^{-27}$ J of a quantum at 5 MHz, which can hardly be called infinitesimal.

A fixed phase shift does not change the frequency. The second example of [1] assigns to a 1° phase shift the new frequency $f_2 = f_1 - f_1/360 = 4{,}986{,}111.11$ Hz. Multiplying a wave by a constant $e^{i\theta}$, or delaying it, leaves its frequency unchanged (Proposition 3.1). A frequency change of $f_1/360$ occurs when the phase *slips* by one degree in every period, that is, when the phase grows at the rate $\dot\theta = 2\pi f_1/360$ (Proposition 3.2). In that case the energy of a quantum changes by

$$E_1 - E_2 = \frac{hf_1}{360} = 9.20\times10^{-30}\ \text{J},$$

and this is the only energy difference of the size suggested by the example. The paper also gives $E_2 = hf_2 \approx 3.313\times10^{-27}$ J; the correct value is 3.304×10^{-27} J, and the difference from E_1 is the number just computed, not 1.848×10^{-27} J.

What hf measures. The energy hf is that of one quantum. The energy of a classical wave is fixed by its amplitude, which the formulas (2.2) do not contain. A loss caused by a passive element is a fraction of the incident energy, whether the wave is classical or quantised, and the quantity to compute is therefore the transmission $|H|^2$, or, per quantum, the probability of absorption.

2.3. What remains

Table I lists the corrections. The kinematic part of [1] stands. The energy part asks a good question in the wrong variables: the phase shift that has a price is the one applied uniformly over a band, the price is a fraction of the energy rather than a multiple of h, and its size is governed by the bandwidth. Sections 4–10 compute it.

TABLE I. The formulas of [1] and their status.

Formula in [1]	Status
$\Phi^\circ = 360^\circ f\Delta t$ and its inversions	correct
$T(\text{deg}) = 1/(360f)$, 555 ps at 5 MHz	correct
"1° phase shift = $T/360$"	angle equated with time; a delay $T/360$ gives 1°
$\Delta E = hf\Delta t$	an action, in J s; its value is 1.83×10^{-36}, not 1.848×10^{-27}
$\Delta E = (2\pi hf_1/360)T(\text{deg})$	an action; equals $2\pi h/360^2$
$\Delta E = (2\pi h/360)\,T(\text{deg})/\Delta t$	an action
$f_2 = f_1 - f_1/360$ after a 1° shift	only if the phase slips 1° per period
$E_2 \approx 3.313\times10^{-27}$ J	3.304×10^{-27} J; $E_1 - E_2 = 9.20\times10^{-30}$ J

3. PHASE, DELAY AND FREQUENCY

3.1. Conventions

A signal is a real function $s(t)$ of finite energy, with Fourier transform $\hat s(\omega) = \int_{\mathbb R} s(t)e^{i\omega t}\,\mathrm dt$, so that $\hat s(-\omega) = \overline{\hat s(\omega)}$. A linear time-invariant system acts by $\widehat{s_{\text{out}}} = H\hat s_{\text{in}}$. With this convention a delay by τ has $H(\omega) = e^{i\omega\tau}$, so a positive phase of H is a *lag*, and causality means that H extends analytically to the upper half-plane $\mathbb C_+ = \{\operatorname{Im} z > 0\}$ (Section 4). The analytic signal of s is

$$\psi(t) = \frac{1}{\pi}\int_0^\infty \hat s(\omega)e^{-i\omega t}\,\mathrm d\omega, \qquad s = \operatorname{Re}\psi.$$

For a narrowband pulse we write $\psi(t) = A(t)e^{-i(\omega_0 t+\phi)}$ with a slowly varying envelope $A \ge 0$, a carrier frequency ω_0 and a carrier–envelope phase ϕ. The phase ϕ locates the crests of the carrier relative to the peak of the envelope; it is the quantity controlled in frequency combs [21, 22].[8]

Three different operations go under the name of a phase shift. We treat them in turn.

3.2. A constant phase factor

The *ideal phase shifter* by θ multiplies every positive frequency by $e^{i\theta}$ and, to keep the signal real, every negative frequency by $e^{-i\theta}$:

$$H_\theta(\omega) = e^{i\theta\,\operatorname{sgn}\omega}, \qquad \psi_{\text{out}} = e^{i\theta}\psi. \tag{3.1}$$

It is not causal.[9]

Proposition 3.1. *The ideal phase shifter preserves the energy $\int s^2\,\mathrm dt$, the power spectrum $|\hat s|^2$ and hence every frequency present in the signal. For a single quantised mode of frequency ω the operator $U = e^{-i\theta N}$, N the number operator, commutes with the Hamiltonian $\hbar\omega(N+\frac12)$, so the probability distribution of the energy is unchanged. If $\sin\theta \neq 0$, the ideal phase shifter is not causal.*

Proof. The first statements follow from $|H_\theta| = 1$ and Parseval's identity, and from $[N,U] = 0$. For the last, $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$, and $i\operatorname{sgn}\omega$ is the transfer function of convolution with the Hilbert kernel $\pm1/(\pi t)$, which does not vanish for $t < 0$. □

[8] In a mode-locked laser ϕ advances by a fixed amount from one pulse to the next; the rate of this slip is the carrier–envelope offset frequency of the comb, which is measured and locked in [21, 22].

[9] With our Fourier convention the multiplier $i\operatorname{sgn}\omega$ has the kernel $1/(\pi t)$, since p. v. $\int \operatorname{sgn}\omega\, e^{-i\omega t}\,\mathrm d\omega = -2i/t$. Hence $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$ has the impulse response $\cos\theta\,\delta(t) + \sin\theta/(\pi t)$, which does not vanish for $t < 0$ unless $\sin\theta = 0$.

So a phase shift, as such, neither changes the frequency nor costs energy. What it costs is causality, and the price of restoring causality in a passive system is the subject of the rest of the paper.

3.3. A delay

A delay τ gives $\psi_{\mathrm{out}}(t) = \psi(t-\tau)$ and the phase $\omega\tau$ at frequency ω, which is the relation (2.1). It is causal and lossless, and it realises any prescribed phase at one frequency. On a band it realises a phase that is proportional to frequency, never a constant one.

3.4. A phase that changes in time

Let a device multiply the analytic signal by $e^{i\theta(t)}$.

Proposition 3.2. *If $\theta(t) = \theta_0 + \Omega t$, then $\widehat{\psi_{\mathrm{out}}}(\omega) = e^{i\theta_0}\hat{\psi}(\omega+\Omega)$: the spectrum is moved down by Ω, and a component of frequency ω leaves with frequency $\omega-\Omega$ and energy $\hbar(\omega-\Omega)$ per quantum. If the device conserves the number of quanta, the energy of the signal changes by the fraction Ω/ω at each frequency, the difference being exchanged with the agent that drives θ.*

Proof. The Fourier transform of $e^{i\Omega t}\psi(t)$ is $\hat{\psi}(\omega+\Omega)$ in our convention. The rest is the relation $E = \hbar\omega$ for each quantum. □

The conservation of the number of quanta is what the Manley–Rowe relations [19] give for a lossless parametric modulator, and the linear phase ramp is the serrodyne frequency translator [20]. A phase that slips by one degree per period has $\Omega = \omega/360$; at 5 MHz this is a shift of 13,888.89 Hz and an energy of $hf/360 = 9.20\times10^{-30}$ J per quantum, the numbers of Section 2. This is the only reading under which a "1° phase shift" changes frequency. It is an active operation, not a passive one.

3.5. Constant phase on a band

The operation that keeps the envelope and moves the carrier is the following.

Proposition 3.3. *Let $0 < a < b$ and let H satisfy $H(-\omega) = \overline{H(\omega)}$ and*

$$H(\omega) = e^{-\alpha(\omega)}e^{i(\theta+\tau\omega)}, \qquad a \le \omega \le b. \tag{3.2}$$

If the spectrum of s lies in $[-b,-a]\cup[a,b]$ and $\alpha = 0$ on $[a,b]$, then $\psi_{\mathrm{out}}(t) = e^{i\theta}\psi(t-\tau)$. In particular a pulse $A(t)e^{-i(\omega_0 t+\phi)}$ leaves as $A(t-\tau)e^{-i(\omega_0(t-\tau)+\phi-\theta)}$: the envelope is delayed by τ and the carrier–envelope phase is moved by $-\theta$. Conversely, if $\psi_{\mathrm{out}} = e^{i\theta}\psi(\cdot-\tau)$ for every such signal, then (3.2) holds with $\alpha = 0$.

Proof. On positive frequencies $\widehat{\psi_{\mathrm{out}}} = H\hat{\psi} = e^{i\theta}e^{i\tau\omega}\hat{\psi}$, which is the transform of $e^{i\theta}\psi(t-\tau)$. The converse follows by testing with signals whose spectra are concentrated near each $\omega \in [a,b]$. □

The phase of H on the band is then $\varphi(\omega) = \theta + \tau\omega$. Its slope τ is the group delay; its intercept

$$\theta = \varphi(\omega) - \omega\varphi'(\omega) = \omega\big(\tau_{\mathrm{ph}} - \tau_{\mathrm{gr}}\big), \qquad \tau_{\mathrm{ph}} = \frac{\varphi}{\omega},\ \tau_{\mathrm{gr}} = \varphi',$$

is the carrier–envelope offset, ω times the excess of phase delay over group delay. A delay has $\theta = 0$. The main theorem says that a passive zero-free system can have $\theta > 0$ only by absorbing, on average over the band, at least $\theta\Lambda$ nepers.

4. PASSIVE SYSTEMS AND HERGLOTZ FUNCTIONS

4.1. Passive systems

Definition 4.1. A *passive system* is a function H analytic in $\mathbb{C}_+$ with $|H(z)| \le 1$ there and with the real symmetry $H(-\bar{z}) = \overline{H(z)}$. Its frequency response is the boundary function $H(\omega) = \lim_{\varepsilon\downarrow0} H(\omega+i\varepsilon)$, which exists for almost every real ω.[10]

The definition encodes three physical requirements. A time-invariant system that never puts out more energy than it receives acts on signals of finite energy by a multiplier of norm at most one; causality makes the multiplier the boundary value of a bounded analytic function in $\mathbb{C}_+$; and a real impulse response gives the symmetry. That these requirements are equivalent to Definition 4.1 is classical [8, 9, 23]. The transmission coefficient of a passive two-port, the reflection coefficient of a passive one-port and the propagation factor of a passive medium (Section 10) are passive systems.

The zeros of a passive system that is not identically zero satisfy the Blaschke condition, and $H = BG$ where B is the Blaschke product of the zeros and G is passive and has no zeros in $\mathbb{C}_+$ [23, Ch. II]. Since the zeros of H come in pairs $\zeta, -\bar{\zeta}$, B can be chosen with the real symmetry.[11] We call H *zero-free* when B is constant. This includes every minimum-phase system, and also the singular inner functions, which are not minimum-phase in the usual sense: the factor $\exp(i\gamma/(\omega_0 - z))$, $\gamma > 0$, is the lossless resonance of a reactance at ω_0 in the sense of Foster [11].

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[10] Fatou's theorem: a bounded analytic function in $\mathbb{C}_+$ has nontangential limits at almost every point of $\mathbb{R}$ [23, Ch. I].

[11] With $B_\zeta(z) = (z-\zeta)/(z-\bar{\zeta})$ one has $B_\zeta(-\bar{z}) = \overline{B_{-\bar{\zeta}}(z)}$, so the product $B_\zeta B_{-\bar{\zeta}}$ has the symmetry, and so does the Blaschke product over such pairs, with the usual unimodular convergence factors chosen in pairs.

4.2. The Herglotz function of a zero-free system

A *Herglotz function* is an analytic map h from $\mathbb{C}_+$ to the closed upper half-plane. Every such function has the representation [9, 16, 23]

$$h(z) = c + \beta z + \int_{\mathbb{R}} \Big(\frac{1}{t - z} - \frac{t}{1 + t^2} \Big) \, \mathrm{d}\mu(t), \tag{4.1}$$

with $c \in \mathbb{R}$, $\beta \geq 0$ and a positive measure μ with $\int \mathrm{d}\mu/(1+t^2) < \infty$. The measure is recovered by the Stieltjes inversion formula,

$$\mu\big((x_1, x_2)\big) + \tfrac{1}{2}\mu(\{x_1, x_2\}) = \lim_{y \downarrow 0} \frac{1}{\pi} \int_{x_1}^{x_2} \operatorname{Im} h(x + iy) \, \mathrm{d}x, \tag{4.2}$$

and $\beta = \lim_{y \to \infty} h(iy)/(iy)$.

Lemma 4.2. *Let H be a zero-free passive system, $H \not\equiv 0$. There are a sign $\sigma \in \{1, -1\}$ and a unique Herglotz function h with*

$$H = \sigma e^{ih}, \qquad h(iy) \in i\mathbb{R} \ (y > 0), \qquad h(-\bar{z}) = -\overline{h(z)}. \tag{4.3}$$

In the representation (4.1) of h one has $c = 0$ and μ is even, so that

$$h(z) = \beta z - \frac{\mu(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2 - z^2} \, \mathrm{d}\mu(t). \tag{4.4}$$

Proof. On the imaginary axis $H(iy) = \overline{H(iy)}$ is real and, having no zeros, of constant sign σ. Since $\mathbb{C}_+$ is simply connected, σH has an analytic logarithm, which we fix to be real on $i\mathbb{R}_+$. Put $h = -i \log(\sigma H)$. Then $\operatorname{Im} h = -\log|H| \geq 0$, so h is a Herglotz function, and $h(iy) \in i\mathbb{R}$. The function $g(z) = -\overline{h(-\bar{z})}$ is analytic, satisfies $e^{ig} = \overline{\sigma H(-\bar{z})} = \sigma H = e^{ih}$, and agrees with h on $i\mathbb{R}_+$; hence $g = h$. In (4.1), $c = \operatorname{Re} h(i) = 0$; the symmetry and (4.2) show that μ is even, and pairing t with $-t$ in (4.1) gives (4.4). Uniqueness holds because two such logarithms differ by a constant in $2\pi\mathbb{Z}$ that vanishes on $i\mathbb{R}_+$. □

We call

$$\varphi(\omega) = \operatorname{Re} h(\omega + i0), \qquad \alpha(\omega) = \operatorname{Im} h(\omega + i0) = -\log|H(\omega)|$$

the *phase* and the *attenuation* of H. Then $\arg H(\omega) = \varphi(\omega) + \frac{1-\sigma}{2}\pi$ modulo 2π. The phase φ is the argument of σH continued from the imaginary axis, where it vanishes; when H is continuous and positive at $\omega = 0$ and nonzero on $[0, b]$, it is simply the argument of H unwrapped from $\varphi(0) = 0$. The representation (4.4) separates the three ingredients of a zero-free system: the delay βz, the lossless resonances, which are the atoms and the singular part of μ, and the losses, which are the absolutely continuous part of μ, with density α/π.

4.3. The band

Fix $0 < a < b$ and write $S = [-b, -a] \cup [a, b]$.

Definition 4.3. A zero-free passive system H is *regular on the band* if there is an open interval $I \supset [a, b]$, $I \subset (0, \infty)$, such that H extends continuously to $\mathbb{C}_+ \cup I$ and has no zeros on I.

Lemma 4.4. *If H is regular on the band, then h extends continuously to $\mathbb{C}_+ \cup I$, φ and α are continuous on I, and the restriction of μ to $I \cup (-I)$ is $\pi^{-1}\alpha(|t|)\,\mathrm{d}t$. In particular μ has no atoms at $\pm a, \pm b$.*

Proof. Near each point of I the function σH is continuous and nonzero, so its logarithm, normalised as in Lemma 4.2, is continuous up to I. The convergence in (4.2) is then uniform on compact subintervals of I, which gives $\mu = \pi^{-1}\alpha \, \mathrm{d}t$ on I; evenness gives the statement on $-I$. □

4.4. Lossless bands

The following is the form of Foster's reactance theorem [11] that we need.

Proposition 4.5. *Let $H = B\sigma e^{ih}$ be passive, with B a real-symmetric Blaschke product whose zeros stay away from I, and let σe^{ih} be regular on the band. If $|H| = 1$ on I, then the argument of H is real-analytic and nondecreasing on I, and strictly increasing unless H is constant.*

Proof. On I we have $\alpha = 0$, so by Lemma 4.4 μ vanishes on I and (4.4) continues analytically across I, with

$$\varphi'(x) = \beta + \frac{\mu(\{0\})}{x^2} + \int_{(0,\infty)\setminus I} \frac{2(t^2 + x^2)}{(t^2 - x^2)^2} \, \mathrm{d}\mu(t) \ \geq \ 0.$$

Each factor $(z - \zeta)/(z - \bar{\zeta})$ of B, $\operatorname{Im}\zeta > 0$, has argument with derivative $2\operatorname{Im}\zeta/|x - \zeta|^2 > 0$ on the real line, and the series for B converges with its derivatives on I. The sum is strictly positive unless $\beta = 0$, $\mu = 0$ and B is constant. □

Corollary 4.6. *Under the hypotheses of Proposition 4.5, a passive system that is lossless on I and has constant argument on $[a, b]$ is a constant, $H \equiv \pm 1$.*

So constant phase on a band is impossible without loss. The main theorem measures how much loss.

5. A TEST FUNCTION FOR THE BAND

5.1. The functions ρ and W

Let

$$\rho(z) = \Big(\frac{z - a}{z - b} \Big)^{1/2} \Big(\frac{z + a}{z + b} \Big)^{1/2}, \qquad z \in \mathbb{C}_+,$$

with principal square roots. The map $z \mapsto (z-a)/(z-b)$ is a real Möbius transformation of determinant $a-b<0$ and sends $\mathbb{C}_+$ onto the lower half-plane; $z \mapsto (z+a)/(z+b)$ has determinant $b-a>0$ and sends $\mathbb{C}_+$ onto itself. Neither meets the negative real axis, so ρ is analytic and zero-free in $\mathbb{C}_+$.[12] Put

$$W(z) = \rho(z) - 2 + \frac{1}{\rho(z)} = \frac{(\rho(z)-1)^2}{\rho(z)}, \tag{5.1}$$

and, for $a<t<b$,

$$r(t) = \sqrt{(t^2-a^2)(b^2-t^2)},$$
$$p(t) = \frac{a^2+b^2-2t^2}{r(t)} = \sqrt{\frac{b^2-t^2}{t^2-a^2}} - \sqrt{\frac{t^2-a^2}{b^2-t^2}}. \tag{5.2}$$

Lemma 5.1. *(i) $\rho(-\bar z) = \overline{\rho(z)}$, and $\rho(z) = 1 + \frac{b^2-a^2}{2z^2} + O(z^{-4})$, so $W(z) = O(z^{-4})$ as $z \to \infty$ in $\overline{\mathbb{C}_+}$.*

(ii) ρ and W extend continuously to $\overline{\mathbb{C}_+}$ minus the four points $\pm a, \pm b$, near which $|W(z)| = O(|z \mp a|^{-1/2})$, $O(|z \mp b|^{-1/2})$.

(iii) On $\mathbb{R}\setminus S$, ρ is real and positive, and $W = (\rho-1)^2/\rho$ is real and strictly positive.

(iv) For $a<t<b$, $\rho(t+i0) = -i\,s(t)$ with $s(t) = \sqrt{(t^2-a^2)/(b^2-t^2)}$, hence

$$W(t) = -2 + i\,p(t), \qquad W(-t) = -2 - i\,p(t).$$

Proof. (i) The symmetry holds for each factor. Since $\rho^2 = (z^2-a^2)/(z^2-b^2) = 1+(b^2-a^2)/(z^2-b^2)$ and $\rho \to 1$, the expansion follows, and $W = (\rho-1)^2/\rho$. (ii) Near b the factor $(z-a)/(z-b)$ behaves like a constant over $z-b$, so $\rho \sim c(z-b)^{-1/2}$ and $W \sim \rho$; near a, $\rho \sim c(z-a)^{1/2}$ and $W \sim 1/\rho$. The points $-a, -b$ follow by symmetry. (iii) For real $t \notin S$ both quotients are positive. (iv) For $a<t<b$ and $\varepsilon \downarrow 0$ the number $(t-a)/(t-b+i\varepsilon)$ has negative real part and negative imaginary part, so its principal square root tends to $-i\sqrt{(t-a)/(b-t)}$, while $\sqrt{(t+a)/(t+b)}$ is positive. Hence $\rho = -is$ and $W = -is - 2 + i/s = -2 + i(1/s - s) = -2 + ip$. The value at $-t$ follows from (i). □

Figure 3 shows $\operatorname{Re} W$: a plateau at -2 over the band, positive values elsewhere, and integrable spikes at the band edges.

5.2. The kernel identity

Lemma 5.2. *For $s>0$, $s \neq a, b$,*

$$\frac{1}{\pi}\,\mathrm{p.v.}\int_a^b p(x)\,\frac{2x}{x^2-s^2}\,\mathrm{d}x = \begin{cases} -2, & a<s<b, \\ W(s), & s \notin [a,b]. \end{cases} \tag{5.3}$$

[12] The principal square root is analytic and zero-free on $\mathbb{C}\setminus(-\infty,0]$, and an open half-plane bounded by the real axis does not meet $(-\infty,0]$.

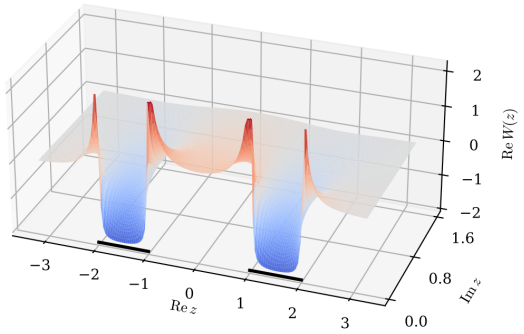


FIG. 3. The real part of the test function W over the upper half-plane for $a=1$, $b=2$ (values clipped at ± 2.2). On the real axis it equals -2 on the band (black segments) and is strictly positive elsewhere.

Proof. Apply Cauchy's theorem to $W(z)/(z-s)$ on the boundary of the half-disc of radius R in $\mathbb{C}_+$, indented by semicircles of radius ε above $s, \pm a, \pm b$. By Lemma 5.1(ii) the small semicircles at $\pm a, \pm b$ contribute $O(\varepsilon^{1/2})$; by Lemma 5.1(i) the large one contributes $O(R^{-4})$; the one at s contributes $-\pi i W(s)$ in the limit, W being continuous at s. Therefore

$$\mathrm{p.v.}\int_{\mathbb{R}} \frac{W(x)}{x-s}\,\mathrm{d}x = \pi i\, W(s).$$

Taking imaginary parts and using that $\operatorname{Im} W$ vanishes off S and equals $\pm p(|x|)$ on $\pm[a,b]$ (Lemma 5.1(iii),(iv)),

$$\pi \operatorname{Re} W(s) = \mathrm{p.v.}\int_a^b p(x)\Big(\frac{1}{x-s} + \frac{1}{x+s}\Big)\,\mathrm{d}x$$
$$= \mathrm{p.v.}\int_a^b p(x)\,\frac{2x}{x^2-s^2}\,\mathrm{d}x.$$

By Lemma 5.1, $\operatorname{Re} W(s) = -2$ for $a<s<b$ and $\operatorname{Re} W(s) = W(s)$ otherwise. □

5.3. Moments of the weight

The weight p is positive on (a,c) and negative on (c,b), where $c^2 = (a^2+b^2)/2$. Under $\sigma = x^2$ it becomes antisymmetric, and this gives all the sign information we need.

Lemma 5.3. *Let $A = a^2$, $B = b^2$ and $\varpi(\sigma) = (A+B-2\sigma)/\sqrt{(\sigma-A)(B-\sigma)}$, so that $x\,p(x)\,\mathrm{d}x = \frac{1}{2}\varpi(x^2)\,\mathrm{d}(x^2)$.*

(i) $\varpi(A+B-\sigma) = -\varpi(\sigma)$, and $\int_a^b x\,p(x)\,\mathrm{d}x = 0$.

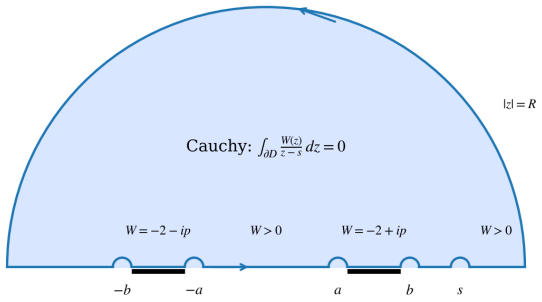


FIG. 4. The contour of the proof of Lemma 5.2: a half-disc of radius R in $\mathbb{C}_+$, indented above the branch points $\pm a, \pm b$ of W and above the pole s. The branch cuts of W are the two bands $\pm[a,b]$ (thick); the indentations at the branch points contribute nothing in the limit, the one at s contributes $-\pi i W(s)$.

(ii) If f is continuous and nonincreasing on $[A,B]$, then $\int_a^b x\,p(x)f(x^2)\,\mathrm{d}x \geq 0$, with equality only for constant f.

(iii) $J := \int_a^b p(x)\,\mathrm{d}x > 0$ *and* $\int_a^b \frac{p(x)}{x}\,\mathrm{d}x = \frac{\pi(b-a)^2}{2ab}$.

(iv) $\int_a^b x^{2j+1}p(x)\,\mathrm{d}x < 0$ *for* $j \geq 1$, *and* $\int_a^b x^3 p(x)\,\mathrm{d}x = -\frac{\pi}{8}(b^2-a^2)^2$.

Proof. (i) The antisymmetry is visible, and it gives $\int_A^B \varpi = 0$. (ii) With $\sigma_m = (A+B)/2$,

$$\int_a^b xpf(x^2)\,\mathrm{d}x = \frac{1}{2}\int_A^{\sigma_m} \varpi(\sigma)\big(f(\sigma) - f(A+B-\sigma)\big)\,\mathrm{d}\sigma,$$

and both factors are nonnegative for $\sigma < \sigma_m$; the integral vanishes only if $f(\sigma) = f(A+B-\sigma)$ throughout, which for a monotone f means f constant. (iii) Apply (ii) to $f(\sigma) = \sigma^{-1/2}$ and $f(\sigma) = \sigma^{-1}$. The value follows from the elementary integrals

$$\int_A^B \frac{\mathrm{d}\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} = \pi,$$
$$\int_A^B \frac{\mathrm{d}\sigma}{\sigma\sqrt{(\sigma-A)(B-\sigma)}} = \frac{\pi}{\sqrt{AB}},$$

which give $\frac{1}{2}\big((A+B)\pi/\sqrt{AB} - 2\pi\big) = \pi(b-a)^2/(2ab)$. (iv) Apply (ii) to $f(\sigma) = -\sigma^j$. For $j=1$ use also $\int_A^B \sigma\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(A+B)/2$ and $\int_A^B \sigma^2\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(3A^2 + 2AB + 3B^2)/8$. □

The first moment vanishing is the reason why delays are free; the positivity of J is the reason why a constant phase lag is not.

6. THE SUM RULE AND THE MAIN THEOREM

6.1. The sum rule

Theorem 6.1. *Let $H = \sigma e^{ih}$ be a zero-free passive system that is regular on the band, with phase φ, attenuation α and Herglotz measure μ. Then*

$$\int_a^b \alpha(\omega)\,\mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\,p(\omega)\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W(t)\,\mathrm{d}\mu(t), \tag{6.1}$$

and every term is finite. The last integral is nonnegative, and it vanishes if and only if $\mu(\mathbb{R}\setminus S) = 0$.

The proof uses one fact about the Hilbert transform $\mathcal{H}f(x) = \pi^{-1}\,\mathrm{p.v.}\int f(t)\,\mathrm{d}t/(x-t)$ on the line, the antisymmetry

$$\int_{\mathbb{R}} f\,\mathcal{H}g\,\mathrm{d}x = -\int_{\mathbb{R}} (\mathcal{H}f)\,g\,\mathrm{d}x, \qquad f \in L^q,\ g \in L^{q'}, \tag{6.2}$$

valid for $1 < q < \infty$ by M. Riesz's theorem [24, 25].[13]

Proof. Finiteness. φ is continuous on $[a,b]$ and p is integrable. On $I\setminus[a,b]$ the measure is $\pi^{-1}\alpha\,\mathrm{d}t$ with α bounded (Lemma 4.4), and W has integrable singularities at a, b. Away from $I\cup(-I)$, W is bounded and $O(t^{-4})$, while $\int \mathrm{d}\mu/(1+t^2) < \infty$.

Decomposition. Choose $a_1 < a$ and $b_1 > b$ with $[a_1,b_1] \subset I$, let $I_1 = (a_1,b_1)$, and split $\mu = \mu_1 + \mu_2$ with μ_1 the restriction of μ to $I_1 \cup (-I_1)$. By (4.4),

$$h = \beta z + h_1 + h_2,$$
$$h_j(z) = -\frac{\mu_j(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu_j(t),$$

and $\varphi = \beta x + \operatorname{Re} h_1 + \operatorname{Re} h_2$ on (a,b).

The delay. $\frac{1}{2}\int_a^b \beta x\,p(x)\,\mathrm{d}x = 0$ by Lemma 5.3(i).

The far part. The measure μ_2 has no mass in $I_1 \cup (-I_1)$, so h_2 is continuous up to (a,b) and the kernel $2x/(t^2-x^2)$ is bounded by a constant times $(1+t^2)^{-1}$ for $x \in [a,b]$ and $t \notin I_1$. Fubini's theorem and Lemmas 5.2 and 5.3(iii) give

$$\frac{1}{2}\int_a^b \operatorname{Re} h_2\,p\,\mathrm{d}x = -\frac{\mu(\{0\})}{2}\int_a^b \frac{p}{x}\,\mathrm{d}x - \frac{\pi}{2}\int_{(0,\infty)\setminus I_1} W\,\mathrm{d}\mu$$
$$= -\frac{\pi}{4}\int_{\mathbb{R}\setminus(I_1\cup -I_1)} W\,\mathrm{d}\mu,$$

where we used $W(0) = (b-a)^2/(ab)$,[14] the evenness of W on $\mathbb{R}\setminus S$ and of μ.

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[13] Riesz's theorem is the boundedness of $\mathcal{H}$ on L^q, $1 < q < \infty$. For smooth compactly supported f, g the antisymmetry is Fubini's theorem for the truncated kernel $\mathbf{1}_{|x-t|>\varepsilon}/(x-t)$, which is odd under $x \leftrightarrow t$; it extends to $L^q \times L^{q'}$ by density and Riesz's bound.

[14] $\rho(0) = \sqrt{a/b}\sqrt{a/b} = a/b$, so $W(0) = a/b - 2 + b/a = (b-a)^2/(ab)$. With Lemma 5.3(iii), $\int_a^b p/x\,\mathrm{d}x = \pi(b-a)^2/(2ab) = \frac{\pi}{2}W(0)$, which turns the atom at 0 into $-\frac{\pi}{4}W(0)\mu(\{0\})$.

The near part. Here $\mu_1 = \psi(t)\,\mathrm{d}t$ with $\psi = \pi^{-1}\alpha(|t|)$ on $I_1 \cup (-I_1)$, a bounded function. For almost every $x \in (a,b)$,

$$\begin{aligned}\operatorname{Re} h_1(x) &= \mathrm{p.\,v.}\int_{I_1} \psi(t)\frac{2x}{t^2-x^2}\,\mathrm{d}t \\ &= \mathrm{p.\,v.}\int_{I_1} \psi(t)\Big(\frac{1}{t-x}-\frac{1}{t+x}\Big)\,\mathrm{d}t,\end{aligned}$$

the boundary value of the real part of a Cauchy integral with bounded density [24, 26]. We claim that

$$\begin{aligned}&\int_a^b p(x)\,\operatorname{Re} h_1(x)\,\mathrm{d}x \\ &\qquad = \int_{I_1}\psi(t)\Big(\mathrm{p.\,v.}\int_a^b p(x)\frac{2x}{t^2-x^2}\,\mathrm{d}x\Big)\,\mathrm{d}t. \quad (6.3)\end{aligned}$$

For the part $1/(t+x)$ of the kernel this is Fubini's theorem, the kernel being bounded. For the part $1/(t-x)$ it is (6.2) with $f = p\,\mathbf{1}_{[a,b]} \in L^{3/2}$ and $g = \psi\mathbf{1}_{I_1} \in L^3$; the function p lies in L^q for every $q<2$ because its singularities are of order $|x-a|^{-1/2}$ and $|x-b|^{-1/2}$. By Lemma 5.2 the inner integral in (6.3) equals 2π for $t \in (a,b)$ and $-\pi W(t)$ for $t \in I_1 \setminus [a,b]$. Hence

$$\begin{aligned}\frac{1}{2}\int_a^b p\,\operatorname{Re} h_1\,\mathrm{d}x &= \pi\int_a^b \psi\,\mathrm{d}t - \frac{\pi}{2}\int_{I_1\setminus[a,b]} W\,\mathrm{d}\mu \\ &= \int_a^b \alpha\,\mathrm{d}t - \frac{\pi}{4}\int_{(I_1\cup -I_1)\setminus S} W\,\mathrm{d}\mu.\end{aligned}$$

Adding the three parts gives (6.1). Since $W>0$ on $\mathbb{R}\setminus S$ and μ has no atoms at $\pm a, \pm b$, the last term vanishes exactly when $\mu(\mathbb{R}\setminus S)=0$. □

The identity (6.1) holds for every phase. Its content is that the band loss is a fixed linear functional of the band phase plus a nonnegative price for everything the system does outside the band: a lossless resonance at $\omega_0 \notin [a,b]$ with weight γ in μ costs $\frac{\pi}{2}W(\omega_0)\gamma$,[15] a pole at the origin costs $\frac{\pi}{4}W(0)\mu(\{0\})$, and absorption outside the band costs its W-weighted total.

6.2. Constant phase lag

Put

$$\begin{aligned}\Lambda &= \frac{J}{2(b-a)} \\ &= \frac{1}{2(b-a)}\int_a^b \frac{a^2+b^2-2\omega^2}{\sqrt{(\omega^2-a^2)(b^2-\omega^2)}}\,\mathrm{d}\omega > 0.\end{aligned} \quad (6.4)$$

Section 7 shows $\Lambda = (K(\kappa)-E(\kappa))/\kappa$.

[15] The measure is even, so the resonance puts mass γ at both $\pm\omega_0$, and $W(-\omega_0)=W(\omega_0)$ off the band; the two atoms give $\frac{\pi}{4}\cdot 2W(\omega_0)\gamma$.

Theorem 6.2 (Main theorem)**.** *Let H be a zero-free passive system, regular on the band $[a,b]$, whose phase on the band is*

$$\varphi(\omega) = \theta + \tau\omega, \qquad a<\omega<b,$$

with $\theta \ge 0$ and $\tau \in \mathbb{R}$. Then

$$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\mathrm{d}\omega \;\ge\; \theta\,\Lambda. \quad (6.5)$$

Equality holds if and only if $\mu(\mathbb{R}\setminus S)=0$; for every $\theta\ge 0$ it is attained by the system $H_\star$ of Theorem 8.2.

Proof. By Lemma 5.3(i), $\frac{1}{2}\int_a^b(\theta+\tau\omega)p\,\mathrm{d}\omega = \frac{1}{2}\theta J = (b-a)\theta\Lambda$, and Theorem 6.1 gives the inequality and the case of equality. □

Three remarks make the statement concrete.

Remark 6.3 (Constant argument). If $\arg H \equiv \theta$ modulo 2π on the band, then $\varphi = \theta - \frac{1-\sigma}{2}\pi$ modulo 2π, and since φ is continuous it is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. The theorem applies with θ' when $\theta' > 0$. Which representative occurs is not decided by the band: it depends on how much phase the system has accumulated from the imaginary axis, for instance from zero frequency.

Remark 6.4 (Lossless resonances). The equality case shows that every lossless resonance outside the band, and every atom of μ at zero frequency, raises the loss strictly above the bound. Shaping the phase with reactances, the classical tool of all-pass and phase-equaliser design, therefore works against a constant phase lag. Figure 5 shows the price of a unit of measure off the band, the factor $\frac{\pi}{4}W(t)$ of (6.1).

Remark 6.5 (Passivity everywhere is needed). The constant-phase element $H(z) = (-iz/a)^{-\gamma}$, $0<\gamma<1$, familiar from dielectric relaxation [18], has phase $\theta = \gamma\pi/2$ at every positive frequency and attenuation $\alpha = \gamma\log(\omega/a) \ge 0$ on $[a,\infty)$. Over an octave its mean attenuation is $(2\theta/\pi)(2\log 2 - 1) = 0.2459\,\theta$, below the bound $0.2735\,\theta$. There is no contradiction: $|H|>1$ for $\omega<a$, so H is not passive. The bound depends on passivity at all frequencies, not only on the band.

6.3. Constant phase lead

For a negative intercept the same identity, applied after an inversion of the frequency axis, gives a weighted bound.

Theorem 6.6. *Let H be as in Theorem 6.2 but with $\theta<0$. Then*

$$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\frac{ab}{\omega^2}\,\mathrm{d}\omega \;\ge\; |\theta|\,\Lambda - \frac{\pi}{4}\,\tau\,(b-a). \quad (6.6)$$

For $\tau=0$ equality holds if and only if $\beta=0$ and μ vanishes on $\mathbb{R}\setminus(S\cup\{0\})$; it is attained by $H_\star(-ab/z)$, where $H_\star$ is the extremal system for $|\theta|$.

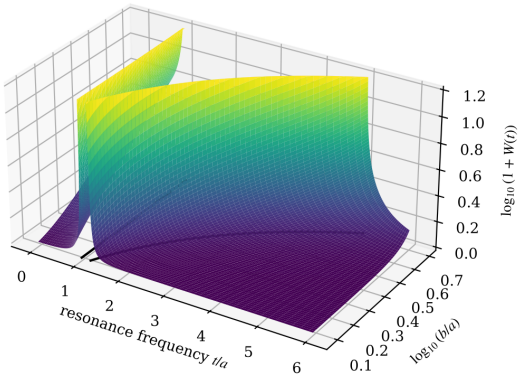


FIG. 5. The cost $\log_{10}(1+W(t))$ of a lossless resonance at frequency t, as a function of t/a and of the band ratio b/a. It vanishes only on the band (the gap between the two sheets), grows without bound at the band edges, and decays like t^{-4} at high frequency (Lemma 5.1(i)); at $t=0$ it is $W(0)=a/b+b/a-2>0$.

Proof. The map $z\mapsto -ab/z$ preserves $\mathbb{C}_+$, so $\tilde h(z)=h(-ab/z)$ is a Herglotz function with the symmetry of (4.3), and $\tilde H=\sigma e^{i\tilde h}$ is zero-free, passive and regular on the band, since $x\mapsto ab/x$ maps a neighbourhood of $[a,b]$ onto another. For $x\in(a,b)$ the point $-ab/(x+i0)$ lies above $-ab/x\in(-b,-a)$, so by the oddness of φ,

$$\tilde\varphi(x)=-\varphi(ab/x),\qquad \tilde\alpha(x)=\alpha(ab/x).$$

Substituting $x=ab/\omega$ and using $r(ab/\omega)=ab\,r(\omega)/\omega^2$,

$$\int_a^b f(ab/x)\,p(x)\,\mathrm{d}x=\int_a^b f(\omega)\,p^*(\omega)\,\mathrm{d}\omega,$$
$$p^*(\omega)=\frac{a^2+b^2-2a^2b^2/\omega^2}{r(\omega)},$$

for every integrable f. With $f=1$ and $f(\omega)=\omega$ this gives $\int_a^b p^*=J$ and $\int_a^b \omega p^*=ab\int_a^b p/x=\pi(b-a)^2/2$ (Lemma 5.3(iii)). Theorem 6.1 for $\tilde H$ now reads

$$\int_a^b \alpha(\omega)\frac{ab}{\omega^2}\,\mathrm{d}\omega=-\frac12\int_a^b(\theta+\tau\omega)p^*\,\mathrm{d}\omega+\frac{\pi}{4}\int_{\mathbb{R}\setminus S}W\,\mathrm{d}\tilde\mu,$$

which is (6.6) after division by $b-a$. The measure $\tilde\mu$ is the image of μ under $t\mapsto -ab/t$ with density ab/t^2 on $\mathbb{R}\setminus\{0\}$, plus an atom $ab\beta$ at 0 coming from the term βz; an atom of μ at 0 becomes a linear term of $\tilde h$ and costs nothing. This gives the case of equality, and $H_\star(-ab/z)$ has $\beta=0$ and its measure on $S\cup\{0\}$. □

6.4. The worst frequency

Corollary 6.7. *Let H be zero-free, passive and regular on the band, with $\arg H\equiv\theta$ modulo 2π on $[a,b]$. Then*

$$\max_{[a,b]}\alpha\ \ge\ \operatorname{dist}(\theta,\pi\mathbb{Z})\,\Lambda,$$

so at some frequency of the band the transmitted power is at most $e^{-2\operatorname{dist}(\theta,\pi\mathbb{Z})\Lambda}$. If the phase θ' of Remark 6.3 is positive, also

$$\exp\Big(\frac{1}{b-a}\int_a^b\log|H(\omega)|^2\,\mathrm{d}\omega\Big)\le e^{-2\theta'\Lambda},$$

that is, the geometric mean of the power transmission over the band is at most $e^{-2\theta'\Lambda}$.

Proof. By Remark 6.3 the phase is a constant $\theta'\equiv\theta\pmod\pi$. If $\theta'>0$ then $\theta'\ge\operatorname{dist}(\theta,\pi\mathbb{Z})$ and Theorem 6.2 bounds the mean, hence the maximum, of α. If $\theta'<0$ then $|\theta'|\ge\operatorname{dist}(\theta,\pi\mathbb{Z})$, and since $\int_a^b ab\,\omega^{-2}\,\mathrm{d}\omega=b-a$ the left side of (6.6) with $\tau=0$ is at most $\max\alpha$. The second statement is (6.5) with $\log|H|=-\alpha$. □

Remark 6.8 (The lead extremal). For the system $H_\star(-ab/z)$ of Theorem 6.6, whose phase is $-|\theta|$ on the band, Theorem 6.1 applied directly gives the plain mean attenuation

$$\frac{1}{b-a}\int_a^b\alpha\,\mathrm{d}\omega=|\theta|\,\frac{E(\kappa)-\kappa'^2K(\kappa)}{\kappa},\qquad \kappa'^2=1-\kappa^2,$$

because its measure off S is the atom $ab\,\tau_\star$ at 0, of cost $\frac{\pi}{4}W(0)ab\tau_\star$, with $\tau_\star$ from Theorem 8.2 and the evaluation of Section 7. For an octave this is $0.2656\,|\theta|$, slightly below $\Lambda=0.2735$. Whether it is the least plain mean among systems with constant phase lead is not known.

7. THE CONSTANT

We use the complete elliptic integrals of modulus $k\in[0,1)$,

$$K(k)=\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}},$$
$$E(k)=\int_0^{\pi/2}\sqrt{1-k^2\sin^2\phi}\,\mathrm{d}\phi,$$

and the moduli attached to the band,

$$k=\sqrt{1-a^2/b^2},\qquad k'=a/b,\qquad \kappa=\frac{b-a}{b+a}=\frac{1-k'}{1+k'}.$$

The facts on K and E that we use are collected in Appendix A.

Proposition 7.1. *With* $I_0 = \int_a^b \mathrm{d}t/r(t)$ *and* $I_2 = \int_a^b t^2\,\mathrm{d}t/r(t)$,

$$I_0 = \frac{K(k)}{b}, \qquad I_2 = b\,E(k),$$

and

$$\begin{aligned} J &= (a^2+b^2)I_0 - 2I_2 = b\big[(1+k'^2)K(k) - 2E(k)\big] \\ &= 2(a+b)\big[K(\kappa) - E(\kappa)\big]. \end{aligned} \tag{7.1}$$

Consequently

$$\Lambda = \frac{K(\kappa) - E(\kappa)}{\kappa}. \tag{7.2}$$

Proof. Substitute $t^2 = b^2 - (b^2-a^2)\sin^2\phi$, $0 \le \phi \le \pi/2$. Then $b^2 - t^2 = (b^2-a^2)\sin^2\phi$, $t^2 - a^2 = (b^2 - a^2)\cos^2\phi$, $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$, hence $\mathrm{d}t/r(t) = -\,\mathrm{d}\phi/t$ and $t = b\sqrt{1-k^2\sin^2\phi}$. This gives I_0 and I_2, and the second form of J. The Landen transformation (Appendix A) states

$$K(k) = (1+\kappa)K(\kappa), \qquad E(k) = (1+k')E(\kappa) - k'K(k).$$

Inserting these,[16] and using $b(1+\kappa) = 2b^2/(a+b)$ and $1+k' = (a+b)/b$,

$$\begin{aligned} J &= b(1+\kappa)(1+k')^2K(\kappa) - 2b(1+k')E(\kappa) \\ &= 2(a+b)\big[K(\kappa) - E(\kappa)\big]. \end{aligned}$$

Dividing by $2(b-a) = 2\kappa(a+b)$ gives (7.2). □

Proposition 7.2. *(i) With* $c_n = \binom{2n}{n}4^{-n}$,

$$\Lambda = \frac{\pi}{2}\sum_{n\ge1}\frac{2n}{2n-1}\,c_n^2\,\kappa^{2n-1} = \frac{\pi}{4}\kappa\Big(1+\frac{3}{8}\kappa^2+\frac{15}{64}\kappa^4+\cdots\Big),$$

a series with positive coefficients;[17] *in particular* Λ *increases with* κ*, that is, with* b/a*.*

(ii) For all $0 < a < b$,

$$\frac{\pi}{4}\,\frac{b-a}{b+a} \;<\; \Lambda \;\le\; \frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2} = \frac{\pi}{16}\,\frac{b^2-a^2}{ab}.$$

(iii) As $a/b \to 0$, $\Lambda = \log\frac{4}{\kappa'} - 1 + o(1) = \frac{1}{2}\log\frac{4b}{a} - 1 + o(1)$, *with* $\kappa' = \sqrt{1-\kappa^2} = 2\sqrt{ab}/(a+b)$.

Proof. (i) From $K = \frac{\pi}{2}\sum c_n^2\kappa^{2n}$ and $E = \frac{\pi}{2}\sum c_n^2\kappa^{2n}/(1-2n)$ one gets $K - E = \frac{\pi}{2}\sum_{n\ge1} c_n^2\kappa^{2n}\,2n/(2n-1)$. The coefficients for $n = 1, 2, 3$ are $\frac{1}{2}$, $\frac{3}{16}$ and $\frac{15}{128}$, which give the displayed expansion after division by κ. (ii) The lower bound is the first term. For the upper bound, the ratio of consecutive coefficients $d_n = c_n^2\,2n/(2n-1)$ is

$$\frac{d_{n+1}}{d_n} = \frac{(2n+1)(2n-1)}{(2n+2)\,2n} < 1,$$

so $d_n \le d_1 = \frac{1}{2}$ and $K - E \le \frac{\pi}{4}\sum_{n\ge1}\kappa^{2n} = \frac{\pi}{4}\kappa^2/(1-\kappa^2)$. Finally $\kappa/(1-\kappa^2) = (b^2-a^2)/(4ab)$. (iii) As $\kappa \to 1$, $K(\kappa) = \log(4/\kappa') + o(1)$ and $E(\kappa) \to 1$ (Appendix A); and $\log(4/\kappa') = \log\big(2(a+b)/\sqrt{ab}\big) = \frac{1}{2}\log(4b/a) + o(1)$. □

For a narrow band with centre $\bar\omega = (a+b)/2$ and width $\Delta\omega = b - a$, $\kappa = \Delta\omega/(2\bar\omega)$, so

$$\Lambda = \frac{\pi}{8}\,\frac{\Delta\omega}{\bar\omega}\Big(1 + O\big((\Delta\omega/\bar\omega)^2\big)\Big). \tag{7.3}$$

The bound is linear in the relative bandwidth, which is what one expects: on an infinitely narrow band a delay achieves any phase without loss. For wide bands it grows only logarithmically in b/a. Table II gives values, also in decibels per degree of phase, which is Λ times $(20/\ln 10)(\pi/180) = 0.15160$, and Figure 6 shows the bound as a surface.

TABLE II. The constant Λ of (7.2), the least mean attenuation per radian of constant phase lag, for several band ratios. The last column is the least worst-frequency power loss for $\theta = 1°$.

b/a	κ	Λ (Np/rad)	dB/deg	$1 - e^{-2\Lambda\pi/180}$
1.01	0.004975	0.003907	0.000592	1.36×10^{-4}
1.1	0.04762	0.03743	0.00567	1.31×10^{-3}
1.5	0.2	0.1595	0.0242	5.55×10^{-3}
2	0.3333	0.2735	0.0415	9.50×10^{-3}
3	0.5	0.4366	0.0662	1.51×10^{-2}
10	0.8182	0.9448	0.143	3.24×10^{-2}
100	0.9802	2.016	0.306	6.80×10^{-2}
1000	0.9980	3.150	0.478	1.04×10^{-1}

8. SHARPNESS: THE EXTREMAL SYSTEMS

8.1. The square root R

Let R be the branch of $\sqrt{(z^2-a^2)(z^2-b^2)}$ that is analytic in $\mathbb{C}\setminus S$ and satisfies $R(z) = z^2\big(1+O(z^{-2})\big)$ at infinity. It exists because a loop around one component of S encircles two branch points.[18]

Lemma 8.1. $R(-z) = R(z)$ *and* $R(\bar z) = \overline{R(z)}$. *On the real axis,* $R(t) = \sqrt{(t^2-a^2)(t^2-b^2)} > 0$ *for* $|t| > b$, $R(t) = -\sqrt{(a^2-t^2)(b^2-t^2)} < 0$ *for* $|t| < a$, *and*

$$R(t+i0) = i\,r(t), \quad R(-t+i0) = -i\,r(t) \qquad (a < t < b).$$

[16] The coefficient of $K(\kappa)$ is $b(1+\kappa)\big[(1+k'^2)+2k'\big] = b(1+\kappa)(1+k')^2$, the term $2k'$ coming from $-2E(k)$.

[17] By (A.1), $K - E = \frac{\pi}{2}\sum_{n\ge1} c_n^2\big(1 - \frac{1}{1-2n}\big)\kappa^{2n}$ and $1 - \frac{1}{1-2n} = \frac{2n}{2n-1}$; dividing by κ gives (i).

[18] Each of the four factors $(z\mp a)^{1/2}$, $(z\mp b)^{1/2}$ changes sign along a loop around its branch point; a loop around $[a,b]$ changes the sign twice, so the product returns to its value.

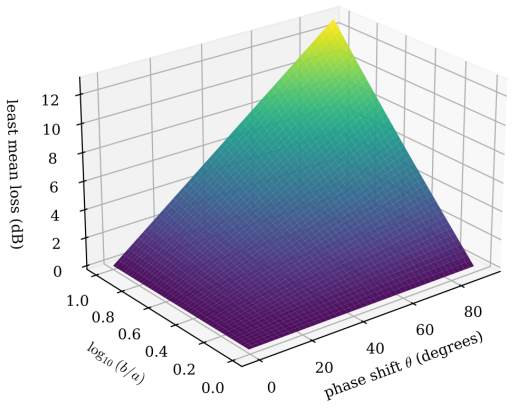


FIG. 6. The least mean loss $\theta\Lambda$, in decibels, over the phase lag θ and the band ratio b/a. The surface is linear in θ and increasing in b/a.

Proof. The symmetries hold because $R(-z)$ and $\overline{R(\bar z)}$ are branches of the same root with the same behaviour at infinity. Near $z=b$, $R(z)=(z-b)^{1/2}G(z)$ with G analytic and $G(b)=\sqrt{2b(b^2-a^2)}>0$, the root being positive for $z>b$. Passing from $t>b$ to $t<b$ through $\mathbb{C}_+$ turns $(z-b)^{1/2}$ into $i\sqrt{b-t}$, so $R=ir$ on (a,b). Near $z=a$ write $R=(z-a)^{1/2}G_1(z)$ with $(z-a)^{1/2}>0$ for $t>a$; then G_1 is positive imaginary on (a,b), and passing to $t<a$ through $\mathbb{C}_+$ gives $R=i\sqrt{a-t}\,G_1<0$. The values on the negative axis follow from the symmetries. □

8.2. The extremal system for constant phase

Put

$$m^2=b^2\,\frac{E(k)}{K(k)},\qquad \tau_\star=\frac{2\theta K(k)}{\pi b}=\frac{4\theta K(\kappa)}{\pi(a+b)},\tag{8.1}$$

so that $a^2<m^2<b^2$, because $m^2=I_2/I_0$ is an average of t^2 over (a,b) (Proposition 7.1), and define

$$h_\star(z)=\tau_\star\int_0^z\frac{s^2-m^2}{R(s)}\,\mathrm{d}s,\qquad H_\star=e^{ih_\star}.\tag{8.2}$$

Theorem 8.2. *Let $\theta>0$.*

(i) The integral (8.2) does not depend on the path in $\mathbb{C}\setminus S$; $h_\star$ is analytic there, odd, real on $\mathbb{R}\setminus S$, and $h_\star(z)=\tau_\star z+O(z^{-1})$ at infinity.

(ii) On the band, $h_\star(x+i0)=\theta+i\alpha_\star(x)$ with

$$\alpha_\star(x)=\tau_\star\int_a^x\frac{m^2-s^2}{r(s)}\,\mathrm{d}s,\tag{8.3}$$

which is positive on (a,b), vanishes at a and b, and is largest at $x=m$.

(iii) $h_\star$ is a Herglotz function, and $H_\star$ is a zero-free passive system, regular on the band, lossless on $\mathbb{R}\setminus S$, with constant phase θ on $[a,b]$ and

$$\frac{1}{b-a}\int_a^b\alpha_\star(\omega)\,\mathrm{d}\omega=\theta\Lambda.$$

Proof. (i) The integrand is even, analytic in $\mathbb{C}\setminus S$, and integrable at the four branch points. A loop around $[a,b]$ can be shrunk onto the two sides of the cut, where $R=\pm ir$; its integral is $\pm2i\int_a^b(s^2-m^2)\,\mathrm{d}s/r(s)=\pm2i(I_2-m^2I_0)=0$. By evenness the same holds around $[-b,-a]$. Hence $h_\star$ is single-valued, and odd. On $(-a,a)$ and on $|t|>b$ the integrand is real, so $h_\star$ is real there; it is also real along the path from 0 to $\pm a$, and the vanishing of the periods shows that it returns to real values beyond $\pm b$. At infinity, $(s^2-m^2)/R(s)=1+O(s^{-2})$, so $h_\star(z)-\tau_\star z$ tends to a constant; it is odd and analytic for $|z|>b$, so its Laurent series has only odd powers and the constant is 0. (ii) For $0<x<a$, $h_\star(x)=\tau_\star\int_0^x(m^2-s^2)\,\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)}$ by Lemma 8.1, and at $x=a$ this is

$$\begin{aligned}&\tau_\star\Big(\frac{m^2}{b}K(k')-b\big(K(k')-E(k')\big)\Big)\\&\quad=\frac{\tau_\star b}{K(k)}\big(E(k)K(k')+E(k')K(k)-K(k)K(k')\big)\\&\quad=\frac{\pi b\,\tau_\star}{2K(k)}=\theta,\end{aligned}$$

by the substitution $s=a\sin\phi$[19] and Legendre's relation (Appendix A). Continuing from a along the upper side of the cut, where $R=ir$,

$$h_\star(x+i0)=\theta+\tau_\star\int_a^x\frac{s^2-m^2}{ir(s)}\,\mathrm{d}s=\theta+i\alpha_\star(x).$$

The integrand of (8.3) is positive for $s<m$ and negative for $s>m$, and its integral over (a,b) is $I_2-m^2I_0=0$; so $\alpha_\star$ rises from 0 at a to its maximum at m and falls back to 0 at b. (iii) The function $v=\operatorname{Im}(h_\star-\tau_\star z)$ is harmonic in $\mathbb{C}_+$, continuous on $\overline{\mathbb{C}_+}$, tends to 0 at infinity by (i), vanishes on $\mathbb{R}\setminus S$, and equals $\alpha_\star\ge0$ on (a,b) and, by the symmetry $h_\star(-\bar z)=-\overline{h_\star(z)}$, also on $(-b,-a)$. By the minimum principle $v\ge0$, so $\operatorname{Im}h_\star\ge\tau_\star\operatorname{Im}z\ge0$. Thus $h_\star$ is a Herglotz function with $h_\star(iy)\in i\mathbb{R}$, $H_\star$ is passive and zero-free, and it is regular on the band since $h_\star$ is continuous up to $(0,\infty)$. As $\operatorname{Im}h_\star=0$ on $\mathbb{R}\setminus S$, (4.2) shows $\mu_\star(\mathbb{R}\setminus S)=0$, and Theorem 6.2 holds with equality. □

[19] It gives $\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)}=\mathrm{d}\phi/(b\sqrt{1-k'^2\sin^2\phi})$, and $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi/\sqrt{1-k'^2\sin^2\phi}=(K(k')-E(k'))/k'^2$ with $a^2/k'^2=b^2$.

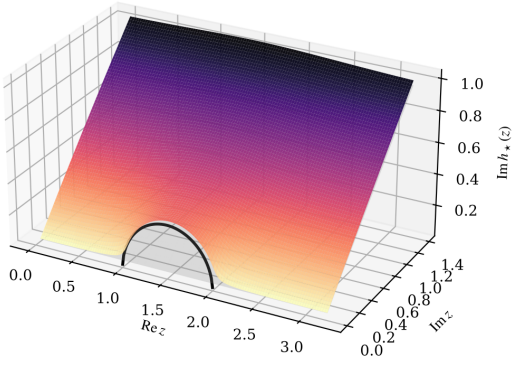


FIG. 7. The surface $\operatorname{Im} h_\star(x+iy)$ over the upper half-plane for an octave and $\theta = 1$. It is positive in $\mathbb{C}_+$, as a Herglotz function must be; on the real axis it vanishes off the band, and over the band its boundary values are the attenuation $\alpha_\star$ (black curve). Far from the band it approaches the plane $\tau_\star y$ (grey) of the delay.

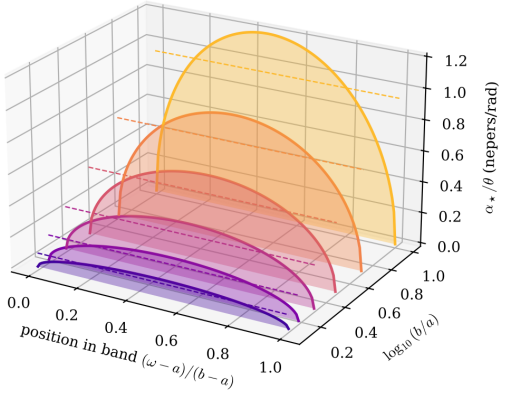


FIG. 8. The attenuation $\alpha_\star/\theta$ of the extremal system across the band, for $b/a = 1.25, 1.5, 2, 3, 5, 10$. Dashed lines mark the mean, Λ. The profile vanishes at both edges.

Remark 8.3. The extremal system is a delay $\tau_\star$ at high frequency, as its asymptotics shows, yet its group delay on the band is zero, its phase being constant there. All of its absorption is in the band, and none of its response outside the band is reactive: in the language of (4.4) it has $\beta = \tau_\star$ and a measure carried by S. For an octave and $\theta = 1$, $\tau_\star = 0.6864/a$, $m = 1.4988\,a$, and the peak attenuation is $\alpha_\star(m) = 0.3482\,\mathrm{Np}$, against the mean 0.2735. Figure 8 shows $\alpha_\star$ for several band ratios, and Figure 7 the function $\operatorname{Im} h_\star$ over $\mathbb{C}_+$.

8.3. Extremal systems for other phases

The construction extends to phases that are not constant, and this is what we need for systems with zeros.

Proposition 8.4. *Let g be continuously differentiable on $[a,b]$ and put*

$$h_g(z) = \frac{2zR(z)}{\pi}\int_a^b \frac{g(t)}{r(t)\,(z^2-t^2)}\,\mathrm{d}t,$$
$$\beta_g = \frac{2}{\pi}\int_a^b \frac{g(t)}{r(t)}\,\mathrm{d}t.$$

(i) h_g is analytic in $\mathbb{C}\setminus S$, odd, real on $\mathbb{R}\setminus S$, continuous up to S from both sides, and $h_g(z) = \beta_g z + O(z^{-1})$ at infinity.

(ii) For $a < x < b$, $\operatorname{Re} h_g(x+i0) = g(x)$ and

$$\operatorname{Im} h_g(x+i0) = v_g(x) = \frac{2x\,r(x)}{\pi}\,\mathrm{p.\,v.}\int_a^b \frac{g(t)}{r(t)\,(x^2-t^2)}\,\mathrm{d}t.$$

(iii) If $v_g \ge 0$ on (a,b) and $\beta_g \ge 0$, then e^{ih_g} is a zero-free passive system, regular on the band, with phase g and attenuation v_g on the band and no measure outside S, so that $\int_a^b v_g = \frac{1}{2}\int_a^b g\,p$.

(iv) For constant $g = \theta$, $h_g = h_\star$.

Proof. (i) The Cauchy integral $q(z) = \int_a^b g(t)\,(r(t)(z^2 - t^2))^{-1}\,\mathrm{d}t$ is analytic off S and even; $zR(z)q(z)$ is odd and real on $\mathbb{R}\setminus S$ because R is real there (Lemma 8.1). The density g/r has singularities of order $|t-a|^{-1/2}$, $|t-b|^{-1/2}$ at the ends, so q is $O(|z-a|^{-1/2})$, $O(|z-b|^{-1/2})$ there and Hölder continuous up to the open cut [26, Ch. 4]; multiplied by R, which vanishes like the square root, the product is continuous up to S. At infinity $q(z) = z^{-2}\int_a^b g/r + O(z^{-4})$ and $zR(z) = z^3(1+O(z^{-2}))$, which gives the expansion. (ii) Write $2z/(z^2-t^2) = 1/(z-t)+1/(z+t)$. By the Plemelj formula, at $x+i0$ with $a<x<b$ the first part of q contributes $\mathrm{p.\,v.}\int g\,(r(t)(x-t))^{-1}\,\mathrm{d}t - i\pi g(x)/r(x)$, so

$$\frac{2}{\pi}xq(x+i0) = \frac{2x}{\pi}\,\mathrm{p.\,v.}\int_a^b \frac{g(t)\,\mathrm{d}t}{r(t)(x^2-t^2)} - \frac{ig(x)}{r(x)}.$$

Multiplying by $R(x+i0) = ir(x)$ gives the real part $g(x)$ and the imaginary part $v_g(x)$. (iii) As in the proof of Theorem 8.2(iii), $\operatorname{Im}(h_g - \beta_g z)$ is harmonic, continuous up to the axis, zero on $\mathbb{R}\setminus S$, equal to $v_g \ge 0$ on S, and tends to 0 at infinity; so $\operatorname{Im} h_g \ge \beta_g \operatorname{Im} z \ge 0$. The last identity is Theorem 6.1. (iv) Let $d = h_g - h_\star$. Both functions have real part θ on the upper side of (a,b), are real on $\mathbb{R}\setminus S$ and satisfy $h(\bar z) = \overline{h(z)}$, so on the lower side the real part is again θ and $d(x \pm i0) = \pm i\,e(x)$ with e real and continuous. Since $R(x \pm i0) = \pm ir(x)$, the function d/R has the same boundary values e/r from both sides of (a,b), and likewise on $(-b,-a)$. It is therefore analytic across

the open cuts, by Morera's theorem, and its singularities at $\pm a, \pm b$ are $O(|z-c|^{-1/2})$, hence removable. So d/R is entire. Both functions have the linear term $\tau_\star z$, because $\beta_\theta = 2\theta I_0/\pi = \tau_\star$, so $d = O(z^{-1})$ and $d/R \to 0$ at infinity. Thus $d = 0$. □

Lemma 8.5. *If $g(t)/t$ is nonincreasing on $[a,b]$, then $v_g \geq 0$. In particular this holds when $g > 0$ and g is nonincreasing, and then also $\beta_g > 0$.*

Proof. Substitute $\sigma = t^2$ and write $G(\sigma) = g(\sqrt{\sigma})/\sqrt{\sigma}$, which is nonincreasing. Then

$$\frac{g(t)\,\mathrm{d}t}{r(t)} = \frac{G(\sigma)\,\mathrm{d}\sigma}{2\sqrt{(\sigma-a^2)(b^2-\sigma)}},$$

and with $\sigma_0 = x^2$,

$$v_g(x) = \frac{x\,r(x)}{\pi} \times \mathrm{p.\,v.}\int_{a^2}^{b^2} \frac{G(\sigma)}{(\sigma_0-\sigma)\sqrt{(\sigma-a^2)(b^2-\sigma)}}\,\mathrm{d}\sigma.$$

Glauert's integral [24],

$$\mathrm{p.\,v.}\int_A^B \frac{\mathrm{d}\sigma}{(\sigma_0-\sigma)\sqrt{(\sigma-A)(B-\sigma)}} = 0$$

for $A < \sigma_0 < B$, which follows from the substitution $\sigma = \frac{A+B}{2} - \frac{B-A}{2}\cos\phi$ and $\mathrm{p.\,v.}\int_0^\pi \mathrm{d}\phi/(\cos\phi - \cos\phi_0) = 0$, allows us to subtract $G(\sigma_0)$:

$$v_g(x) = \frac{x\,r(x)}{\pi}\int_{a^2}^{b^2} \frac{G(\sigma)-G(\sigma_0)}{\sigma_0-\sigma}\,\frac{\mathrm{d}\sigma}{\sqrt{(\sigma-a^2)(b^2-\sigma)}}.$$

For a nonincreasing G the difference quotient is nonnegative, so $v_g \geq 0$. If $g > 0$ is nonincreasing, g/t is decreasing, and $\beta_g > 0$ is clear. □

9. SYSTEMS WITH ZEROS

The main theorem concerns zero-free systems. A passive system may also have zeros in $\mathbb{C}_+$, and a zero contributes phase without contributing loss on the real axis. This section shows that zeros do help, and by how much one zero helps.

9.1. The bound with an inner factor

Proposition 9.1. *Let $H = BG$ be passive, where B is a real-symmetric Blaschke product whose zeros stay away from $I \cup (-I)$ and G is zero-free and regular on the band. Let φ be a continuous argument of H on $[a,b]$ and let ψ be the continuous argument of B on $[a,b]$ for which $\varphi - \psi$ is the phase of G in the sense of Lemma 4.2. Then $\alpha = \alpha_G$ on the band, and*

$$\begin{aligned}\int_a^b \alpha\,\mathrm{d}\omega &= \frac{1}{2}\int_a^b (\varphi-\psi)\,p\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\mu_G \\ &\geq \frac{1}{2}\int_a^b (\varphi-\psi)\,p\,\mathrm{d}\omega.\end{aligned} \tag{9.1}$$

The function ψ is real-analytic and strictly increasing on $[a,b]$.

Proof. The Blaschke product is analytic across I and unimodular there, so $\log|H| = \log|G|$ on the band, and $\arg H = \arg B + \arg G$ modulo 2π. The continuous arguments on $[a,b]$ therefore differ by a constant in $2\pi\mathbb{Z}$, which fixes ψ. Theorem 6.1 applied to G gives (9.1). The monotonicity of ψ is the computation in the proof of Proposition 4.5.[20] □

An increasing ψ behaves like a delay distributed over frequency. A pure delay is free (Lemma 5.3(i)), but a phase that rises steeply at low frequencies and flattens on the band is not a delay: its weighted integral $\int \psi p$ is positive, and (9.1) then allows less loss than (6.5). The simplest instance is a single zero on the imaginary axis.

9.2. One zero

For $\eta > 0$ let

$$B_\eta(z) = \frac{z-i\eta}{z+i\eta}.$$

It is passive, unimodular on $\mathbb{R}$, has the real symmetry, and the argument of $-B_\eta$ on $(0,\infty)$ is

$$\arg\big(-B_\eta(\omega)\big) = \pi - 2\arctan\frac{\eta}{\omega} = 2\arctan\frac{\omega}{\eta} \in (0,\pi). \tag{9.2}$$

For $0 < \theta < \pi$ put $\eta_\theta = b\cot(\theta/2)$, so that $2\arctan(b/\eta_\theta) = \theta$.

Proposition 9.2. *Let $0 < \theta < \pi$ and $\eta \geq \eta_\theta$, and put*

$$g_\eta(\omega) = \theta - 2\arctan\frac{\omega}{\eta}, \qquad H_\eta = -B_\eta\, e^{ih_{g_\eta}},$$

with h_g as in Proposition 8.4. Then H_η is passive and regular on the band, has exactly one zero in $\mathbb{C}_+$, at $i\eta$, has argument θ on $[a,b]$, and

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha_\eta\,\mathrm{d}\omega &= \theta\Lambda - \delta(\eta), \\ \delta(\eta) &= \frac{1}{b-a}\int_a^b \arctan\frac{\omega}{\eta}\,p(\omega)\,\mathrm{d}\omega.\end{aligned} \tag{9.3}$$

[20] For a single factor $(z-\zeta)/(z-\bar\zeta)$ with $\zeta = \xi + i\eta$, $\eta > 0$, the argument on the real line is $\arg(x-\zeta) - \arg(x-\bar\zeta)$, and $\frac{\mathrm{d}}{\mathrm{d}x}\arg(x-\xi\mp i\eta) = \pm\eta/((x-\xi)^2+\eta^2)$; the difference is $2\eta/|x-\zeta|^2$.

The reduction δ is strictly positive and strictly decreasing in η, and

$$\delta(\eta) = \frac{\pi(b^2-a^2)^2}{24(b-a)\eta^3} + O(\eta^{-5}) \qquad (\eta \to \infty). \tag{9.4}$$

Proof. On $[a,b]$ the function g_η is decreasing, and $g_\eta(b) \geq 0$ because $\eta \geq \eta_\theta$. Hence $g_\eta \geq 0$, $g_\eta > 0$ on $[a,b)$, and $g_\eta(t)/t$ is decreasing. By Lemma 8.5, $v_{g_\eta} \geq 0$ and $\beta_{g_\eta} > 0$, so by Proposition 8.4(iii) $e^{ih_{g_\eta}}$ is a zero-free passive system, regular on the band, with phase g_η and attenuation v_{g_η}, and with no measure outside S. Multiplying by $-B_\eta$ keeps passivity and the real symmetry, adds the single zero $i\eta$, leaves the modulus on $\mathbb{R}$ unchanged, and adds the argument (9.2). The argument of H_η on the band is therefore $g_\eta + 2\arctan(\omega/\eta) = \theta$, and Proposition 8.4(iii) gives

$$\int_a^b \alpha_\eta = \frac{1}{2}\int_a^b g_\eta\, p = \frac{\theta}{2}J - \int_a^b \arctan\frac{\omega}{\eta}\, p\,\mathrm{d}\omega,$$

which is (9.3) since $J = 2(b-a)\Lambda$.

For the sign, write $\arctan(x/\eta) = x\, f(x^2)$ with $f(\sigma) = \arctan(\sqrt{\sigma}/\eta)/\sqrt{\sigma}$. Since $u \mapsto \arctan(u)/u$ is strictly decreasing on $(0,\infty)$,[21] f is strictly decreasing, and Lemma 5.3(ii) gives $\delta(\eta) > 0$. Similarly

$$\delta'(\eta) = -\frac{1}{b-a}\int_a^b \frac{x\,p(x)}{\eta^2+x^2}\,\mathrm{d}x < 0,$$

by Lemma 5.3(ii) with the strictly decreasing $f(\sigma) = 1/(\eta^2+\sigma)$. Finally $\arctan u = u - u^3/3 + O(u^5)$ uniformly on $[0, b/\eta]$, and Lemma 5.3(i),(iv) give $\int_a^b x\,p = 0$ and $\int_a^b x^3 p = -\frac{\pi}{8}(b^2-a^2)^2$, which yields (9.4). □

Since δ decreases, the best choice within the family is $\eta = \eta_\theta$, where the phase g_η of the zero-free factor reaches zero at the upper band edge. Figure 9 shows the mean loss of H_η, relative to $\theta\Lambda$, over θ and η/η_θ.

9.3. One zero beats every zero-free system

The statement of Theorem 6.2 fixes the continuous phase φ. A measurement only gives the argument modulo 2π, and a zero-free system with $\arg H \equiv \theta$ may realise θ through a lead $\theta - \pi$, by Remark 6.3. The next corollary takes this into account.

Corollary 9.3. *Let $0 < \theta < a\pi/(a+b)$. Among zero-free passive systems regular on the band with $\arg H \equiv \theta$ modulo 2π on $[a,b]$, the least mean attenuation over the band is $\theta\Lambda$, attained by $H_\star$. The system H_{η_θ} of Proposition 9.2, which has one zero, has the same argument and a strictly smaller mean attenuation.*

[21] Its derivative is $\big(u/(1+u^2) - \arctan u\big)/u^2$, and $\arctan u > u/(1+u^2)$ for $u > 0$ because both vanish at 0 and $\frac{\mathrm{d}}{\mathrm{d}u}\arctan u = \frac{1}{1+u^2} > \frac{1-u^2}{(1+u^2)^2} = \frac{\mathrm{d}}{\mathrm{d}u}\frac{u}{1+u^2}$.

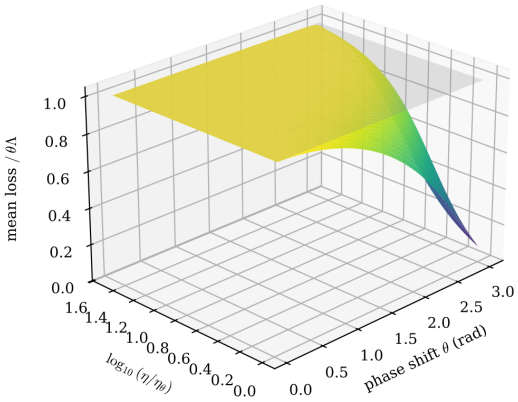


FIG. 9. The mean attenuation of the one-zero system H_η of Proposition 9.2, divided by the zero-free bound $\theta\Lambda$, for $a = 1$, $b = 2$, over the phase θ and $\log_{10}(\eta/\eta_\theta)$. The grey plane is the level 1. The ratio is below 1 everywhere and tends to 1 as the zero moves to infinity.

TABLE III. Mean attenuation over the octave $[1,2]$ of three systems with argument θ modulo 2π: the lag extremal $H_\star$ (the least value among zero-free systems with phase θ), the lead extremal $-H_\star(-ab/z)$ built for $\pi - \theta$, and the one-zero system H_{η_θ}. For $\theta < \pi/3$ the first column is the least value among all zero-free systems (Corollary 9.3).

θ	$\theta\Lambda$	$(\pi-\theta)\,0.26560$	H_{η_θ}
0.5	0.13677	0.70160	0.13443
1	0.27353	0.56880	0.25394
$\pi/2$	0.42966	0.41720	0.34659
2	0.54707	0.30320	0.35977
2.5	0.68383	0.17040	0.27677

Proof. By Remark 6.3 the phase of such a system is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. If $\theta' \geq \theta$, Theorem 6.2 gives mean attenuation at least $\theta'\Lambda \geq \theta\Lambda$. Otherwise $\theta' \leq \theta - \pi < 0$, and Theorem 6.6 with $\tau = 0$ bounds the ab/ω^2-weighted mean by $|\theta'|\,\Lambda \geq (\pi-\theta)\Lambda$. Since $ab/\omega^2 \leq b/a$ on $[a,b]$, the plain mean is at least $(a/b)(\pi-\theta)\Lambda$, which exceeds $\theta\Lambda$ exactly when $\theta < a\pi/(a+b)$. The value $\theta\Lambda$ is attained by $H_\star$ (Theorem 8.2), and Proposition 9.2 gives the second statement. □

For larger θ the lead branch competes. The system $-H_\star(-ab/z)$, built from the extremal for $\pi - \theta$, is zero-free, has $\sigma = -1$ and phase $\theta - \pi$, hence argument θ modulo 2π, and by Remark 6.8 its mean attenuation is $(\pi-\theta)\,(E(\kappa) - \kappa'^2 K(\kappa))/\kappa$. Table III compares the three systems for an octave.

The one-zero system is the best of the three for $\theta \leq \pi/2$; by $\theta = 2$ the lead branch is better. For $\theta = 0.5$ the zero lowers the loss by 1.7%, for $\theta = 1$ by 7.2%.

9.4. An open problem

Problem 9.4. Determine

$$\Lambda_{\mathrm{all}}(\theta) = \inf\Big\{ \frac{1}{b-a}\int_a^b \alpha \,\mathrm{d}\omega \Big\},$$

the infimum over all passive systems H, with or without zeros, that are continuous and nonzero near the band and satisfy $\arg H \equiv \theta$ modulo 2π on $[a,b]$. Is $\Lambda_{\mathrm{all}}(\theta) > 0$ for $\theta \notin \pi\mathbb{Z}$?

Corollary 4.6 shows that the infimum is not attained by a lossless system, but it does not exclude that it is zero. By (9.1), the question is how large $\int_a^b \psi\, p$ can be made by the argument ψ of a Blaschke product, while the remaining phase $\theta - \psi$ is still the phase of a zero-free passive system. The difficulty is the second condition: by Proposition 8.4 it requires $v_g \geq 0$ and $\beta_g \geq 0$ for $g = \theta - \psi$, and $\beta_g \geq 0$ fails as soon as g is too negative on the band.[22]

10. WAVES IN PASSIVE MEDIA

10.1. A slab of a passive medium

Consider a plane wave crossing a slab of thickness L of a linear, homogeneous, isotropic and temporally dispersive medium with relative permittivity $\epsilon(\omega)$ and permeability $\mu_r(\omega)$. We consider the propagation factor of a single pass and ignore reflections at the faces, which is exact for an impedance-matched slab and otherwise describes the field inside the medium. In our convention the factor is

$$H(\omega) = \exp\Big(\frac{i\omega n(\omega) L}{c}\Big), \qquad n = \sqrt{\epsilon\mu_r} = n_r + i n_i.$$

Passivity of the medium means that $z\epsilon(z)$ and $z\mu_r(z)$ are Herglotz functions, real on $i\mathbb{R}_+$ and tending to z as $z \to i\infty$ [9, 15, 16].[23]

Lemma 10.1. *$h(z) = zn(z)L/c$ is a Herglotz function with $h(iy) \in i\mathbb{R}$ and $h(-\bar z) = -\overline{h(z)}$, and its linear coefficient in (4.4) is $\beta = L/c$. Hence $H = e^{ih}$ is a zero-free passive system with $\sigma = 1$, phase $\varphi = \omega n_r L/c$ and attenuation $\alpha = \omega n_i L/c$.*

Proof. Put $f(z) = z\epsilon(z)$ and $g(z) = z\mu_r(z)$, so that $zn(z) = \sqrt{f(z)g(z)}$. For $z \in \mathbb{C}_+$, $\arg f$ and $\arg g$ lie in $[0,\pi]$; we take the square root with argument $\frac12(\arg f + \arg g) \in [0,\pi]$, which is analytic and is the branch with $n \to 1$. So $zn(z)$ maps $\mathbb{C}_+$ to the closed upper half-plane. On $i\mathbb{R}_+$, f and g lie on $i\mathbb{R}_+$, so their product is negative and zn lies on $i\mathbb{R}_+$. The symmetry follows from that of f and g, and $\beta = \lim h(iy)/(iy) = L/c$ because $n(iy) \to 1$. □

The coefficient $\beta = L/c$ is the vacuum delay: no signal crosses the slab faster than light. It drops out of the bound, as every delay does.

Theorem 10.2. *Let the medium be passive, with n continuous and nonzero on a neighbourhood of $[a,b]$ in $\overline{\mathbb{C}_+}$, and suppose that on the band*

$$n_r(\omega) = n_0 + \frac{C}{\omega}, \qquad a \leq \omega \leq b, \tag{10.1}$$

with constants $n_0 \in \mathbb{R}$ and $C \geq 0$. Then

$$\frac{1}{b-a}\int_a^b \omega\, n_i(\omega)\,\mathrm{d}\omega \;\geq\; C\,\Lambda, \tag{10.2}$$

with equality if and only if the Herglotz measure of $zn(z)$ is carried by S.

Proof. By Lemma 10.1, H is zero-free, passive and regular on the band, with phase $\varphi = \omega n_r L/c = CL/c + (n_0 L/c)\,\omega$. Theorem 6.2 with $\theta = CL/c$ and $\tau = n_0 L/c$ gives $\frac{1}{b-a}\int \omega n_i L/c \geq (CL/c)\Lambda$, and L/c cancels. □

The thickness of the slab has dropped out: (10.2) is a property of the material. The condition (10.1) says that the phase index exceeds the group index $n_g = \mathrm{d}(\omega n_r)/\mathrm{d}\omega = n_0$ by C/ω, or equivalently that $\mathrm{d}n_r/\mathrm{d}\omega = -C/\omega^2 < 0$. So Theorem 10.2 is a quantitative form of the old observation that anomalous dispersion requires absorption [6, 7, 9]: an index falling like C/ω across a band requires the frequency-weighted extinction ωn_i to be at least $C\Lambda$ on average. In terms of the intensity absorption coefficient $a_{\mathrm{abs}} = 2\omega n_i/c$ the bound reads $\overline{a_{\mathrm{abs}}} \geq 2C\Lambda/c$.

10.2. Approximate constant phase

The sum rule is linear in the phase, so the bound is stable under perturbations. Put $P = \int_a^b |p(\omega)|\,\mathrm{d}\omega$.

Corollary 10.3. *Let H be zero-free, passive and regular on the band, and suppose $|\varphi(\omega) - \theta - \tau\omega| \leq \varepsilon$ on $[a,b]$. Then*

$$\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega \;\geq\; \theta\Lambda - \frac{P}{2(b-a)}\,\varepsilon.$$

In particular a zero-free system that is lossless on the band cannot approximate a constant phase lag $\theta > 0$ to within $\varepsilon < \theta J/P$.

---

[22] For instance, two zeros chosen so that $g < 0$ on the whole band give $\beta_g = \frac{2}{\pi}\int g/r < 0$; the resulting function e^{ih_g} grows like $e^{-\beta_g y}$ along $i\mathbb{R}_+$ and is not passive.

[23] Analyticity in $\mathbb{C}_+$ is causality of the constitutive relations; $\mathrm{Im}(\omega\epsilon(\omega)) \geq 0$ for real ω is the absence of gain; the limit expresses that every medium is transparent at very high frequency. The Herglotz property in $\mathbb{C}_+$ follows from these boundary properties by the Poisson representation of the harmonic function $\mathrm{Im}(z\epsilon(z))$.

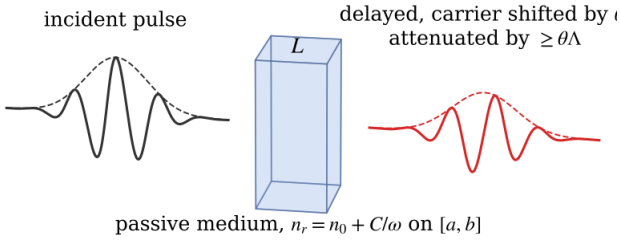


FIG. 10. Schematic of Theorem 10.2. A pulse crosses a slab of a passive medium whose index falls like $n_0 + C/\omega$ on the band. The envelope is delayed by $n_0 L/c$, the carrier is shifted under it by $\theta = CL/c$, and the pulse leaves attenuated (the factor is exaggerated). By Theorem 10.2 the mean of ωn_i over the band is at least $C\Lambda$, whatever the thickness L.

Proof. By Theorem 6.1 and Lemma 5.3(i), $\int_a^b \alpha \geq \frac{1}{2}\int_a^b \varphi p = \frac{1}{2}\theta J + \frac{1}{2}\int_a^b (\varphi - \theta - \tau\omega)p$, and the last term is at least $-\frac{1}{2}\varepsilon P$. If $\alpha = 0$ on the band, then $0 \geq \frac{1}{2}(\theta J - \varepsilon P)$. □

For an octave $J/P = 0.2584$, so a lossless zero-free system cannot hold its phase within $0.258\,\theta$ of any $\theta + \tau\omega$ with $\theta > 0$ across the octave; for the decade $[1, 10]$ the ratio is 0.617. The ratio J/P does not depend on the scale of the band.

Remark 10.4 (Broadband phase shifters). Broadband "90° phase shifters" are pairs of lossless all-pass networks whose phases differ by approximately $\pi/2$ across a band [13]. Each phase is increasing (Proposition 4.5); only their difference is nearly constant. This is consistent with Corollary 10.3: what the theorem forbids is a lossless network whose own phase is nearly constant on the band, and a pair of networks never needs one. A device that shifts a single signal by θ without a reference channel must absorb.

10.3. What the bound means for energy

On the band the power transmission is $T(\omega) = |H(\omega)|^2 = e^{-2\alpha(\omega)}$.

Proposition 10.5. *Let H be as in Theorem 6.2 with $\theta > 0$.*

(i) There is $\omega_1 \in [a, b]$ with $T(\omega_1) \leq e^{-2\theta\Lambda}$, and for every $\varepsilon > 0$ there is a signal with spectrum in $[-b, -a] \cup [a, b]$ that loses at least the fraction $1 - e^{-2\theta\Lambda} - \varepsilon$ of its energy.

(ii) The geometric mean of T over the band is at most $e^{-2\theta\Lambda}$.

(iii) In a quantised description, a single quantum of frequency ω_1 is absorbed with probability at least $1 - e^{-2\theta\Lambda}$, and the mean energy it deposits is at least $(1 - e^{-2\theta\Lambda})\hbar\omega_1$.

(iv) For a narrow band of relative width $\Delta f/f$ and θ in degrees, $1 - e^{-2\theta\Lambda} = \frac{\pi^2}{720}\,\theta\,\frac{\Delta f}{f}\,\big(1 + O(\theta\Delta f/f) + O((\Delta f/f)^2)\big)$.

Proof. (i) The mean of the continuous function α is at least $\theta\Lambda$, so $\alpha(\omega_1) \geq \theta\Lambda$ at some ω_1. A signal whose spectrum is concentrated near $\pm\omega_1$ transmits a fraction of its energy arbitrarily close to $T(\omega_1)$ by Parseval's identity and the continuity of T. (ii) is Corollary 6.7. (iii) A linear lossy element acts on a single quantum as a beam splitter coupling it to the absorbing degrees of freedom; the quantum is transmitted with probability $T(\omega_1)$. (iv) By (7.3), $\Lambda = \frac{\pi}{8}\frac{\Delta f}{f}(1 + O((\Delta f/f)^2))$, and θ degrees are $\pi\theta/180$ radians, so $2\theta\Lambda = \frac{\pi^2}{720}\theta\frac{\Delta f}{f}(1 + \cdots)$; finally $1 - e^{-x} = x(1 + O(x))$. □

The energy that is lost is not the "energy of the phase shift". A phase factor is energy neutral (Proposition 3.1). The loss is the price that causality and passivity attach to producing that factor on a band.

10.4. The 5 MHz example done correctly

Take the example of [1]: a 1° phase shift of a wave at 5 MHz, and give it a band, 4.5 to 5.5 MHz, so $\Delta f/f = 0.2$. Then $\kappa = 0.1$ and

$$\Lambda = 0.078836,$$
$$\theta\Lambda = \frac{\pi}{180}\Lambda = 1.3760 \times 10^{-3}\,\mathrm{Np} = 0.011951\,\mathrm{dB}.$$

A passive device that shifts the carrier–envelope phase by 1° across this band therefore absorbs at least 0.275% of the power at its worst frequency, $1 - e^{-2\theta\Lambda} = 2.748 \times 10^{-3}$, against 2.742×10^{-3} from Proposition 10.5(iv). A quantum of $hf = 3.313 \times 10^{-27}$ J at that frequency deposits on average at least 9.10×10^{-30} J. This happens to be close to the value $hf/360 = 9.20 \times 10^{-30}$ J that a phase slipping by 1° per period would remove (Proposition 3.2), but the agreement is accidental: the ratio of the two numbers is 0.989, close to $\frac{\pi^2}{2}\frac{\Delta f}{f} = 0.987$ by Proposition 10.5(iv), and it tends to zero with the bandwidth. For a single sinusoid, $\Delta f = 0$ and a delay of 555.6 ps does the job without loss.

11. DISCUSSION AND OPEN PROBLEMS

11.1. What has been shown

A passive system can delay a band of frequencies for free, but it cannot shift the phase of the whole band by a constant without absorbing. The least mean absorption is $\theta\Lambda$, with $\Lambda = (K(\kappa) - E(\kappa))/\kappa$ an explicit function of the band ratio, and the bound is attained. The proof is a single sum rule, (6.1), built from the test function W, and every other statement of the paper is a consequence of it or of the explicit extremal system: the lead bound, the worst-frequency and geometric-mean bounds, the bound for anomalous dispersion in a medium, and the stability under perturbations of the phase.

The ingredients are classical: the Herglotz representation, Kramers–Kronig relations, a Cauchy integral, and an elliptic integral reduced by Landen's transformation. What is new, as far as we can tell, is the inequality and its sharpness, and the identification of the price of a constant phase with a single number Λ.

11.2. Open problems

(1) *Systems with zeros.* Determine the infimum of Problem 9.4. Even its positivity is open. A first step would be the best single zero, at an arbitrary point of $\mathbb{C}_+$ rather than on the imaginary axis, and the best pair of zeros $\zeta, -\bar{\zeta}$.

(2) *The plain mean for a lead.* Theorem 6.6 is sharp for the weighted mean. Is the system $H_\star(-ab/z)$ of Remark 6.8 optimal for the plain mean, with value $|\theta|\,(E - \kappa'^2 K)/\kappa$?

(3) *The worst frequency.* Corollary 6.7 bounds $\max\alpha$ by the mean. The sharp bound on $\max\alpha$ is an extremal problem of Chebyshev type: minimise $\sup_{[a,b]} \operatorname{Im} h$ among Herglotz functions with $\operatorname{Re} h = \theta$ on the band. The extremal $H_\star$ has $\max\alpha_\star/\theta\Lambda = 1.273$ for an octave, so the sharp constant lies between Λ and 1.273Λ there.

(4) *Several bands.* For a phase that is constant on each of several disjoint bands, the test function is built from a hyperelliptic square root, and the constant becomes a period of a hyperelliptic differential. We expect the sum rule to carry over with the same proof.

(5) *Reflection and matching.* For reflection coefficients, the Fano bound [12] and the present bound constrain different parts of the same function. A joint bound on the modulus and the argument of a reflection coefficient on a band would combine the two.

(6) *Formalisation.* The finite arithmetic of the paper is checked in Lean (Appendix D). The analytic part, the sum rule and the evaluation of Λ, would require a formalisation of boundary values of Cauchy integrals, of the M. Riesz theorem and of the complete elliptic integrals with Landen's transformation. We have not done this, and the analytic theorems of the paper are not machine-checked.

Appendix A: Complete elliptic integrals

This appendix derives every property of K and E used in the paper. The modulus is $k \in [0,1)$ and the complementary modulus is $k' = \sqrt{1-k^2}$; we write $K' = K(k')$ and $E' = E(k')$.

1. Power series

For $|u| < 1$, $(1-u)^{-1/2} = \sum_{n\ge0} c_n u^n$ and $(1-u)^{1/2} = \sum_{n\ge0} c_n u^n/(1-2n)$, with $c_n = \binom{2n}{n}4^{-n}$; the first is the binomial series, $\binom{-1/2}{n}(-1)^n = c_n$, and the second follows by integrating the first term by term. The Wallis integral $\frac{2}{\pi}\int_0^{\pi/2}\sin^{2n}\phi\,\mathrm{d}\phi = c_n$ follows by induction from the reduction formula $\int_0^{\pi/2}\sin^{2n} = \frac{2n-1}{2n}\int_0^{\pi/2}\sin^{2n-2}$ and the recurrence $c_n = \frac{2n-1}{2n}c_{n-1}$. With $u = k^2\sin^2\phi$ and uniform convergence for $k < 1$,

$$K(k) = \frac{\pi}{2}\sum_{n\ge0} c_n^2 k^{2n}, \qquad E(k) = \frac{\pi}{2}\sum_{n\ge0}\frac{c_n^2}{1-2n}k^{2n}. \tag{A.1}$$

Subtracting, $K - E = \frac{\pi}{2}\sum_{n\ge1} c_n^2\frac{2n}{2n-1}k^{2n}$, which is the series of Proposition 7.2(i).[24]

2. Derivatives

Differentiating under the integral sign,

$$\frac{\mathrm{d}E}{\mathrm{d}k} = \frac{E-K}{k}, \qquad \frac{\mathrm{d}K}{\mathrm{d}k} = \frac{E - k'^2K}{k\,k'^2}. \tag{A.2}$$

The first is immediate: $\frac{\mathrm{d}}{\mathrm{d}k}\sqrt{1-k^2\sin^2\phi} = -k\sin^2\phi/\sqrt{\cdot}$, and $-k^2\sin^2\phi = (1-k^2\sin^2\phi) - 1$. For the second, the derivative is $\int_0^{\pi/2} k\sin^2\phi\,(1-k^2\sin^2\phi)^{-3/2}\,\mathrm{d}\phi$, and the identity

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\frac{\sin\phi\cos\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{1}{k^2}\Big(\sqrt{1-k^2\sin^2\phi} - \frac{k'^2}{(1-k^2\sin^2\phi)^{3/2}}\Big),$$

[24] The coefficients $d_n = c_n^2\,2n/(2n-1)$ are $\frac{1}{2}, \frac{3}{16}, \frac{15}{128}, \frac{175}{2048}, \ldots$; the ratio $d_{n+1}/d_n = (2n+1)(2n-1)/((2n+2)2n)$ follows from $c_{n+1}/c_n = (2n+1)/(2n+2)$.

integrated over $[0,\pi/2]$, gives $\int(1-k^2\sin^2)^{-3/2}=E/k'^2$; then $k^2\sin^2\phi(1-k^2\sin^2\phi)^{-3/2}=(1-k^2\sin^2)^{-3/2}-(1-k^2\sin^2)^{-1/2}$ yields (A.2).[25]

3. Legendre's relation

Lemma A.1. *$E(k)K'+E'K(k)-K(k)K'=\pi/2$ for $0<k<1$.*

Proof. Let $L(k)$ be the left side. Since $\mathrm{d}k'/\mathrm{d}k=-k/k'$, (A.2) applied at k' gives

$$\frac{\mathrm{d}E'}{\mathrm{d}k}=\frac{k(K'-E')}{k'^2},\qquad \frac{\mathrm{d}K'}{\mathrm{d}k}=-\frac{E'-k^2K'}{kk'^2}.$$

Multiply $\mathrm{d}L/\mathrm{d}k$ by kk'^2 and insert the four derivatives:

$$\begin{aligned}kk'^2L'&=k'^2(E-K)K'-E(E'-k^2K')+k^2K(K'-E')\\&\quad+E'(E-k'^2K)-K'(E-k'^2K)+K(E'-k^2K').\end{aligned}$$

Expanding, the terms in KK' and in EE' cancel in pairs, and what remains is

$$kk'^2L'=(k^2+k'^2-1)\,(EK'-KE')=0.$$

So L is constant. As $k\to0$, $K,E\to\pi/2$ and $E'\to1$, while $K-E=O(k^2)$ by (A.1) and $K'=O(\log(1/k))$ by (A.3) below; hence $L=E'K+(E-K)K'\to\pi/2$. □

4. The arithmetic–geometric mean and Landen's transformation

For $p,q>0$ let

$$I(p,q)=\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{p^2\cos^2\phi+q^2\sin^2\phi}},$$

so that $K(k)=I(1,k')$.

Lemma A.2 (Gauss)**.** *$I(p,q)=I\bigl(\frac{p+q}{2},\sqrt{pq}\bigr)$. Consequently $I(p,q)=\pi/(2M(p,q))$, with M the arithmetic–geometric mean of p and q.*

Proof. With $x=q\tan\phi$, $I(p,q)=\frac12\int_{\mathbb R}\bigl((p^2+x^2)(q^2+x^2)\bigr)^{-1/2}\,\mathrm{d}x$. Substitute $x=\frac12(t-pq/t)$, $t>0$, which maps $(0,\infty)$ increasingly onto $\mathbb R$, with $\mathrm{d}x=\frac12(1+pq/t^2)\,\mathrm{d}t$. Then $4x^2+(p+q)^2=(t^2+p^2)(t^2+q^2)/t^2$ and $x^2+pq=\frac14(t+pq/t)^2$, so

$$\bigl((p_1^2+x^2)(q_1^2+x^2)\bigr)^{-1/2}\,\mathrm{d}x=\frac{\mathrm{d}t}{\sqrt{(t^2+p^2)(t^2+q^2)}},$$

with $p_1=\frac{p+q}{2}$, $q_1=\sqrt{pq}$. Integrating over $t\in(0,\infty)$ gives $I(p_1,q_1)=\frac12\int_{\mathbb R}((p^2+t^2)(q^2+t^2))^{-1/2}\,\mathrm{d}t=I(p,q)$. The iterates converge to $M(p,q)$, and $I(M,M)=\pi/(2M)$. □

Proposition A.3 (Landen)**.** *Let $0<k'<1$ and $\kappa=(1-k')/(1+k')$. Then*

$$K(k)=(1+\kappa)K(\kappa),\qquad E(k)=(1+k')E(\kappa)-k'K(k).$$

Proof. By Lemma A.2, $K(k)=I(1,k')=I\bigl(\frac{1+k'}{2},\sqrt{k'}\bigr)=\frac{2}{1+k'}I\bigl(1,\frac{2\sqrt{k'}}{1+k'}\bigr)$, and $\frac{2\sqrt{k'}}{1+k'}=\sqrt{1-\kappa^2}=\kappa'$, so $K(k)=\frac{2}{1+k'}K(\kappa)=(1+\kappa)K(\kappa)$. For E, regard k as a function of κ, $k=2\sqrt\kappa/(1+\kappa)$, $k'=(1-\kappa)/(1+\kappa)$, and differentiate the relation for K with (A.2). Since

$$\frac{\mathrm{d}k}{\mathrm{d}\kappa}=\frac{1-\kappa}{\sqrt\kappa(1+\kappa)^2},\qquad kk'^2=\frac{2\sqrt\kappa(1-\kappa)^2}{(1+\kappa)^3},$$
$$\text{so}\qquad\frac{\mathrm{d}k/\mathrm{d}\kappa}{kk'^2}=\frac{1+\kappa}{2\kappa(1-\kappa)},$$

the derivative of $K(k)$ is $\bigl(E(k)-k'^2K(k)\bigr)\frac{1+\kappa}{2\kappa(1-\kappa)}$, and that of $(1+\kappa)K(\kappa)$ is $K(\kappa)+\bigl(E(\kappa)-(1-\kappa^2)K(\kappa)\bigr)/(\kappa(1-\kappa))$. Equating, multiplying by $2\kappa(1-\kappa)/(1+\kappa)$ and using $k'^2(1+\kappa)=(1-\kappa)^2/(1+\kappa)$,

$$\begin{aligned}E(k)&=\frac{2E(\kappa)}{1+\kappa}+\frac{K(\kappa)}{1+\kappa}\Bigl((1-\kappa)^2+2\kappa(1-\kappa)\\&\qquad-2(1-\kappa)(1+\kappa)\Bigr)\\&=\frac{2E(\kappa)}{1+\kappa}-(1-\kappa)K(\kappa).\end{aligned}$$

Since $2/(1+\kappa)=1+k'$ and $(1-\kappa)K(\kappa)=k'K(k)$, this is the claim. □

5. Behaviour at the ends

As $k\to0$, (A.1) gives $K,E=\frac\pi2+O(k^2)$. As $k\to1$, $E(k)\to1$ and

$$K(k)=\log\frac{4}{k'}+O\bigl(k'^2\log k'\bigr).\tag{A.3}$$

The limit of E is dominated convergence. For K, the substitution $x=k'\tan\phi$ of Lemma A.2 gives $K(k)=\int_0^\infty\bigl((1+x^2)(k'^2+x^2)\bigr)^{-1/2}\,\mathrm{d}x$. Split at $x=1$:

$$\int_0^1\frac{\mathrm{d}x}{\sqrt{k'^2+x^2}}=\operatorname{arsinh}\frac{1}{k'}=\log\frac{2}{k'}+o(1),$$

and, by dominated convergence,

$$\int_0^1\frac{(1+x^2)^{-1/2}-1}{\sqrt{k'^2+x^2}}\,\mathrm{d}x\to\log\frac{2}{1+\sqrt2},$$
$$\int_1^\infty\frac{\mathrm{d}x}{\sqrt{(1+x^2)(k'^2+x^2)}}\to\log(1+\sqrt2),$$

[25] To check the displayed identity, put $s=\sin\phi$, $c=\cos\phi$, $\Delta=1-k^2s^2$. The derivative of $sc\,\Delta^{-1/2}$ is $(c^2-s^2)\Delta^{-1/2}+k^2s^2c^2\Delta^{-3/2}=\Delta^{-3/2}\bigl((c^2-s^2)\Delta+k^2s^2c^2\bigr)$, and with $c^2=1-s^2$ the bracket is $1-2s^2+k^2s^4=\frac{1}{k^2}\bigl(\Delta^2-k'^2\bigr)$.

the first limit being $\int_0^1 \big(\frac{1}{x\sqrt{1+x^2}} - \frac{1}{x}\big)\,\mathrm{d}x = \log\frac{2}{1+\sqrt{2}}$, both from the primitive $-\operatorname{arsinh}(1/x)$ of $1/(x\sqrt{1+x^2})$. The three limits add up to $\log(4/k')$. The error term is [27, 19.12.1].

Appendix B: The elementary integrals

1. Moments of the arcsine weight

For $A < B$ put $m = \frac{A+B}{2}$, $d = \frac{B-A}{2}$ and $\sigma = m - d\cos\phi$, $0 \le \phi \le \pi$. Then $(\sigma - A)(B - \sigma) = d^2\sin^2\phi$ and $\mathrm{d}\sigma = d\sin\phi\,\mathrm{d}\phi$, so $\mathrm{d}\sigma/\sqrt{(\sigma-A)(B-\sigma)} = \mathrm{d}\phi$, and

$$\begin{aligned}
\int_A^B \frac{\mathrm{d}\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} &= \pi,\\
\int_A^B \frac{\sigma\,\mathrm{d}\sigma}{\sqrt{\cdot}} &= \pi m = \frac{\pi}{2}(A+B),\\
\int_A^B \frac{\sigma^2\,\mathrm{d}\sigma}{\sqrt{\cdot}} &= \pi\Big(m^2 + \frac{d^2}{2}\Big)\\
&= \frac{\pi}{8}\big(3A^2 + 2AB + 3B^2\big),\\
\int_A^B \frac{\mathrm{d}\sigma}{\sigma\sqrt{\cdot}} &= \int_0^\pi \frac{\mathrm{d}\phi}{m - d\cos\phi}\\
&= \frac{\pi}{\sqrt{m^2-d^2}} = \frac{\pi}{\sqrt{AB}}.
\end{aligned}$$

The last integral is the classical $\int_0^\pi \mathrm{d}\phi/(m - d\cos\phi) = \pi/\sqrt{m^2-d^2}$ for $m > |d|$, which follows from $u = \tan(\phi/2)$.[26]

These give Lemma 5.3(iii),(iv): with $x\,p(x)\,\mathrm{d}x = \frac{1}{2}\varpi(\sigma)\,\mathrm{d}\sigma$ and $p(x)\,\mathrm{d}x/x = \frac{1}{2}\varpi(\sigma)\,\mathrm{d}\sigma/\sigma$,

$$\begin{aligned}
\int_a^b \frac{p}{x}\,\mathrm{d}x &= \frac{1}{2}\Big((A+B)\frac{\pi}{\sqrt{AB}} - 2\pi\Big)\\
&= \frac{\pi(b-a)^2}{2ab},\\
\int_a^b x^3 p\,\mathrm{d}x &= \frac{1}{2}\Big(\frac{\pi}{2}(A+B)^2\\
&\qquad - \frac{\pi}{4}(3A^2+2AB+3B^2)\Big)\\
&= -\frac{\pi}{8}(B-A)^2.
\end{aligned}$$

2. The integrals I_0, I_2 and J

With $t^2 = b^2 - (b^2-a^2)\sin^2\phi$ (Proposition 7.1), $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$ and $r(t) = (b^2-a^2)\sin\phi\cos\phi$, so $\mathrm{d}t/r = -\,\mathrm{d}\phi/t$ and

$$\begin{aligned}
I_0 &= \int_0^{\pi/2} \frac{\mathrm{d}\phi}{b\sqrt{1-k^2\sin^2\phi}} = \frac{K(k)}{b},\\
I_2 &= \int_0^{\pi/2} b\sqrt{1-k^2\sin^2\phi}\,\mathrm{d}\phi = bE(k).
\end{aligned}$$

Hence $J = (a^2+b^2)I_0 - 2I_2 = b\big((1+k'^2)K(k) - 2E(k)\big)$. By Proposition A.3, $-2E(k) = -2(1+k')E(\kappa) + 2k'K(k)$, and $(1+k'^2) + 2k' = (1+k')^2$, so

$$\begin{aligned}
J &= b(1+k')^2K(k) - 2b(1+k')E(\kappa)\\
&= b(1+k')^2(1+\kappa)K(\kappa) - 2b(1+k')E(\kappa).
\end{aligned}$$

With $1+k' = (a+b)/b$ and $1+\kappa = 2b/(a+b)$ the coefficients are $2(a+b)$ and $2(a+b)$, which is (7.1).

3. The integrals on $(0,a)$

With $s = a\sin\phi$,

$$\sqrt{(a^2-s^2)(b^2-s^2)} = a\cos\phi\,b\sqrt{1-k'^2\sin^2\phi},$$

so

$$\begin{aligned}
\int_0^a \frac{\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= \frac{K(k')}{b},\\
\int_0^a \frac{s^2\,\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= b\big(K(k') - E(k')\big),
\end{aligned}$$

the second because $a^2\sin^2\phi = b^2k'^2\sin^2\phi$ and

$$\int_0^{\pi/2} \frac{k'^2\sin^2\phi\,\mathrm{d}\phi}{\sqrt{1-k'^2\sin^2\phi}} = K(k') - E(k').$$

These are the integrals in the proof of Theorem 8.2(ii), where

$$\frac{m^2}{b}K' - b(K'-E') = \frac{b}{K}\big(EK' - KK' + E'K\big) = \frac{\pi b}{2K}$$

by $m^2 = b^2E/K$ and Lemma A.1.

Appendix C: The extremal profile in closed form

1. Three formulas for $\alpha_\star$

Lemma C.1. *For $a < x < b$,*

$$\begin{aligned}
\alpha_\star(x) &= \tau_\star\int_a^x \frac{m^2-s^2}{r(s)}\,\mathrm{d}s\\
&= \frac{2x\,r(x)\theta}{\pi}\,\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x^2-t^2)}\\
&= \frac{2\theta\,r(x)}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}.
\end{aligned}$$

[26] With $u = \tan(\phi/2)$, $\cos\phi = (1-u^2)/(1+u^2)$ and $\mathrm{d}\phi = 2\,\mathrm{d}u/(1+u^2)$, so the integral is $\int_0^\infty 2\,\mathrm{d}u/\big((m-d)+(m+d)u^2\big) = \pi/\sqrt{(m-d)(m+d)}$.

The last form shows at once that $\alpha_\star > 0$ on (a,b) and that it vanishes like $\sqrt{x-a}$ and $\sqrt{b-x}$ at the band edges.

Proof. The first equality is Theorem 8.2(ii), the second Proposition 8.4(ii),(iv) with $g=\theta$. For the third, apply Cauchy's formula to $1/R$, which is analytic off S, $O(z^{-2})$ at infinity, and has the jumps $1/R(t+i0)-1/R(t-i0) = -2i/r(t)$ on (a,b) and $+2i/r(t)$ on $(-b,-a)$ by Lemma 8.1. Then

$$\frac{1}{R(z)} = \frac{1}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)}\Big(\frac{1}{z-t}-\frac{1}{z+t}\Big).$$

At $z=x+i0$ the left side is $-i/r(x)$, purely imaginary, so the real part of the right side vanishes: $\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x-t)} = \int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}$. Since $\frac{2x}{x^2-t^2} = \frac{1}{x-t}+\frac{1}{x+t}$, the second form equals the third. □

2. The mean of $\alpha_\star$

By Fubini's theorem applied to the first form,

$$\begin{aligned}\int_a^b \alpha_\star\,\mathrm{d}x &= \tau_\star\int_a^b \frac{(m^2-s^2)(b-s)}{r(s)}\,\mathrm{d}s\\ &= \tau_\star\big(b(m^2I_0-I_2)-m^2I_1+I_3\big),\end{aligned}$$

with $I_j=\int_a^b s^j\,\mathrm{d}s/r(s)$. The first bracket vanishes by the choice of m. With $\sigma=s^2$, $I_1=\frac{\pi}{2}$ and $I_3=\frac{\pi}{4}(a^2+b^2)$ by Appendix B, so

$$\begin{aligned}\int_a^b \alpha_\star &= \tau_\star\frac{\pi}{4}\big(a^2+b^2-2m^2\big)\\ &= \frac{\theta K(k)}{2b}\big(a^2+b^2\big)-\theta b\,E(k) = \frac{\theta}{2}J,\end{aligned}$$

in agreement with Theorem 6.2.[27]

3. The lead extremal

The system $H_\star(-ab/z)$ of Theorem 6.6 has, in the representation (4.4), linear coefficient 0, measure on S given by its attenuation, and an atom $ab\,\tau_\star$ at the origin (Theorem 6.6, proof). Its phase on the band is $-|\theta|$. The sum rule (6.1), with the atom costing $\frac{\pi}{4}W(0)\,ab\,\tau_\star$ and $W(0)=(b-a)^2/(ab)$, gives

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega &= -|\theta|\,\Lambda+\frac{\pi}{4}(b-a)\tau_\star\\ &= -|\theta|\,\frac{K-E}{\kappa}+|\theta|\,\kappa K\\ &= |\theta|\,\frac{E(\kappa)-\kappa'^2K(\kappa)}{\kappa},\end{aligned}$$

where K,E are taken at κ and $\frac{\pi}{4}(b-a)\tau_\star = |\theta|\,(b-a)K(\kappa)/(a+b) = |\theta|\,\kappa K(\kappa)$. This is Remark 6.8.

[27] Here $\tau_\star\pi/4 = \theta K(k)/(2b)$ and $\tau_\star\pi m^2/2 = \theta b E(k)$; the last expression is $\frac{\theta}{2}((a^2+b^2)I_0-2I_2)$.

Appendix D: Verification

No statement of the paper rests on a computation: every result has a proof in the text. Two kinds of checks accompany the source files.

Numerical checks. The script `anc/checks.py` re-evaluates the identities of the paper in high-precision arithmetic: the boundary values of W and the kernel identity (5.3) at sample points; the moments of Lemma 5.3; the three forms of J and the Landen relations; the bounds on Λ and Table II; m^2, Legendre's relation and $\tau_\star$; the three formulas of Lemma C.1 and the mean of $\alpha_\star$; the octave values of Remark 8.3; the lead extremal; the single-zero reduction, its asymptotics and Table III; the ratios J/P of Corollary 10.3; and the numbers of Sections 2 and 10. It reports 120 checks passed and none failed.

Lean. The file `lean/PhaseShiftCost.lean` is checked by the kernel of Lean 4 (version 4.34.0, core library only). It contains 57 theorems and no `sorry`. It verifies the dimensional analysis and the arithmetic of Section 2; the coefficients d_n of the series of Λ, their identification with $\binom{2n}{n}^2 4^{-2n}\,2n/(2n-1)$ for $n\le 60$, and their monotonicity for every n; enclosures of Λ/π to nine decimals for six band ratios, from partial sums of the series and the coefficient bound behind the tail estimate; and the polynomial identities used in Sections 5 to 7 and 9 and in the appendices. Of the 57 theorems, 29 depend on no axiom, one on `propext` only, and the others on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), through the tactics `grind` and `omega`.

The analytic theorems are not formalised. The sum rule (Theorem 6.1), the identification of Λ with elliptic integrals and with its series, the extremal system, and the results on zeros rest on boundary values of Cauchy integrals, the M. Riesz theorem, Landen's transformation and the summation of a geometric series, none of which is formalised in the file. They are proved in the text and in Appendices A to C.

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Data availability. The LaTeX source, the script that re-evaluates every numerical value of the paper (120 checks) and the Lean 4 certificate of its finite and algebraic parts (57 theorems, no `sorry`) are archived in the repository

https://github.com/creelie/gsc, release v1.0.0. No statement of the paper depends on them. Anthropic Claude assisted with exposition, figures, the numerical checks and the Lean certificate; the authors checked every proof and are responsible for the text.

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